TECHNICAL PROPOSAL

AI-Driven Recruitment Assistant

RFQ No. OIA/5302/2026

Submitted to

Oman Investment Authority (OIA)

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Submitted by

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2 Executive Summary



The joint venture of Modern Arkan Al Thiba SPC (owned by Mr Hamoud Al Busaidi) and UIS International LLC (an international company, http://www.uisco.net, with a subsidiary in Oman) who prides itself with a 46.2% Omanization rate and thirty percent Omani ownership (Mr Hamoud Al Busaidi) have joined forces to deliver international best practices and proven products locally in Oman. This project builds the AI-Driven Recruitment assistant for the Oman Investment Authority(OIA), integrated with the Jadarah application, using proven in-house AI capabilities delivered by UIS International ("UIS Cloud"). UIS Cloud brings some of the brightest minds in the world from the most elite universities, with over two decades of experience developing and deploying AI systems in natural language processing, machine learning, multilingual document understanding, and conversational and dialogue systems, the foundational capabilities on which the five modules of this offering are built. The solution comprises AI-driven CV parsing, smart candidate matching, standardised assessment exams, an AI Interviewer with a customizable avatar, and application scoring and recommendation.

Business Analysis and Workflow Discovery

Objective: Conduct a comprehensive business analysis to understand OIA's operational workflows, business processes, and specific integration requirements for the AI-Driven Recruitment Assistant.

UIS Cloud understands the depth of business analysis this engagement requires and will assign Meredith Andrews as our dedicated Business Analyst, on the ground with the project for the full duration of the engagement. Meredith will work directly with OIA stakeholders across HR, IT, and relevant operational units to map current recruitment workflows end-to-end, document business processes and integration touchpoints with the Jadarah application, and translate the findings into structured requirement artefacts that feed into technical design and implementation. Meredith remains engaged through implementation to validate that delivery matches the captured requirements.

Modules with Model Recommendations

For modules such as CV Parsing, Smart Candidate Matching, and Application Scoring, we have proposed specific models drawn from our AI team's reference stack. These areas have well-established methodologies and named model families that are widely applied to structured extraction, retrieval and scoring in bilingual environments. For instance, in CV Parsing, PaddleOCR-VL-1.6 is used for layout-aware document parsing, with PP-OCRv5 Arabic providing a deterministic Arabic text-recognition path across multi-format resumes in Arabic and English. Similarly, multilingual embeddings with reranking are standard in candidate-matching systems, and interpretable dimensional scoring against versioned weights and rules is well established for structured application scoring. By naming these models, we anchor the solution in proven technologies while preserving the flexibility to substitute components that meet the same acceptance criteria.

Modules with a Confirmed Reference Stack

Across the five modules, UIS Cloud has proposed a specific reference stack rather than deferring model selection to the implementation phase. Any component may be replaced only if the substitute meets the same accuracy, latency, fairness, security and offline-licensing criteria, with the final configuration confirmed during the Weeks 1-2 verification sprint against the agreed gates.

For CV Parsing, Smart Candidate Matching, and Application Scoring, the stack uses PaddleOCR-VL-1.6 with PP-OCRv5 Arabic for layout-aware and deterministic Arabic document parsing, integrated into a modular pipeline of native text extraction, schema-constrained extraction, deterministic validation and calibrated confidence with candidate confirmation of low-confidence fields. Smart Candidate Matching and Application Scoring share a single matching component that combines an OIA-governed skills and titles taxonomy, deterministic eligibility rules, bilingual lexical search, and multilingual embeddings with reranking; scores are calculated by versioned dimensions and weights and are recalculable independently, with matched evcy rules shown to recruiters before shortlist confirmation.

For the AI Interviewer and the Assessment Exams platform, the AI Interviewer combines NVIDIA Nemotron 3.5 ASR Streaming for Arabic and English speech recognition, Qwen3-14B for controlled bilingual dialogue (with Jais-2-8B-Chat as the Arabic-focused alternative), SILMA-TTS with diacritisation and an OIA pronunciation lexicon for Arabic speech synthesis, a deterministic phoneme-to-viseme avatar renderer, and dedicated audio, video and language feature extractors for behavioural assessment. The Assessment Exams platform ships with a committed 950-item authored bank rendered in Arabic and English as 1,900 language versions with native-speaker equivalence review, a full proctoring stack, and gVisor-class or Kata Containers-class sandboxing for live coding.

Commitment to Utilizing Latest Models and Best Practices

With a confirmed reference stack across all five modules, our commitment is to keep that stack current and evidence-led through the life of the platform. We will continuously evaluate emerging models and integrate improved versions where they meet the same acceptance criteria that governed the initial selection. Our approach includes:

Continuous Evaluation: Testing emerging models across speech, bilingual language, document understanding, matching, and behavioural analytics against the agreed benchmarks before any promotion.

Controlled Substitution: Replacing any component only when the substitute meets or improves on accuracy, latency, fairness, security, and offline-licensing criteria, with every change flowing through the model registry and release gates.

Governance and Human Oversight: Maintaining the release gate as the sole route for model, rule, weight and question-bank changes, running continuous performance and drift monitoring through the support period, and confirming every AI output through authorised human review.

The result is a platform that stays governed, auditable, and aligned to OIA's operational and compliance requirements throughout the contract lifecycle.

2.1 Project Overview & Goals

The engagement targets seven outcomes for OIA:

Accelerate initial candidate screening by automating CV extraction and profile creation from multi-format documents in Arabic and English, so that candidate volume no longer sets the pace of screening.

Improve shortlist quality and defensibility through evidence-backed matching that combines an OIA-governed skills and titles taxonomy with bilingual retrieval and reranking, and shows recruiters the specific CV passages that support each score.

Standardise candidate assessment across cognitive, behavioural and discipline-specific competencies with a committed 950-item authored bank rendered in Arabic and English as 1,900 language versions.

Scale interviewing capacity to run concurrent, structured, bilingual interviews with dynamic follow-up under a controlled dialogue framework, preserving fairness and evidence traceability at any candidate volume.

Return explainable rankings and application scores to Jadarah, with recommendations a recruiter can accept, adjust or reject, and candidate-facing derivations a candidate can question and receive an answer from the record.

Sustain fairness, accessibility and audit reconstruction across every AI-influenced decision, in line with the MTCIT AI Regulatory Framework and WCAG 2.0 AA.

Ensure independent operation after handover by transferring the full technical and operational asset (source code, documentation, question banks, taxonomies, weights and operating know-how) so that OIA can operate, extend and audit the platform without supplier dependence.

2.2 Proposed Solution

The solution comprises five modules operating on the shared foundation described above. Each module produces evidence-linked outputs that another module can consume and a recruiter can confirm.

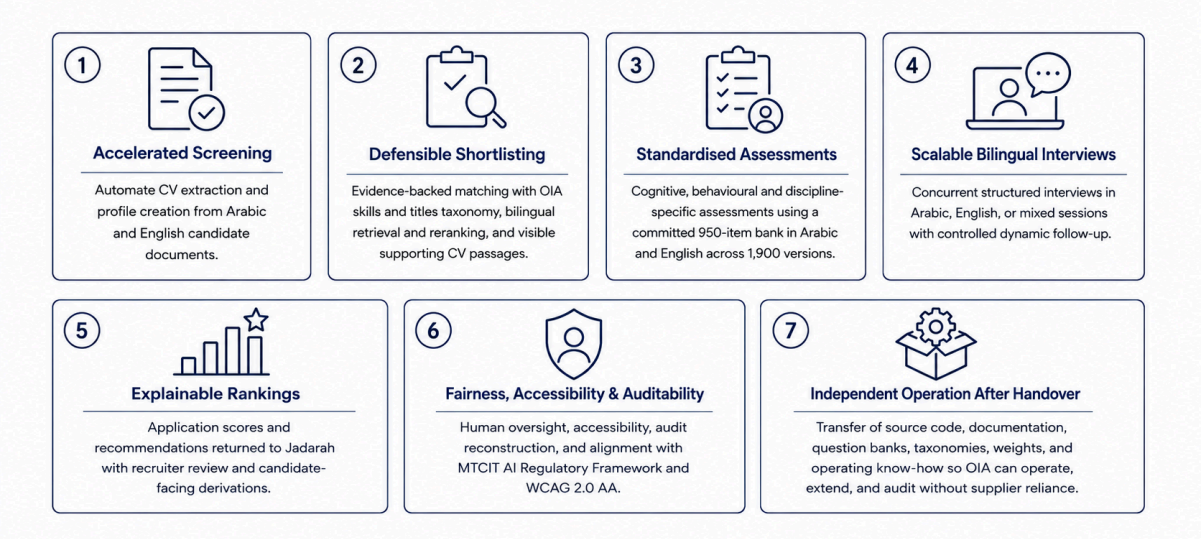
Module 1 - CV Parsing: Automatic extraction of structured candidate data from resumes in PDF, Word and text documents, including scanned and image-based files, in Arabic and English. The pipeline combines native text extraction, layout-aware OCR, schema-constrained extraction, deterministic validation and calibrated confidence. Candidates review parsed data through a step-by-step interface, correct or add fields, and confirm before the profile is written back to Jadarah. When a candidate re-uploads, the system produces a differential update rather than an overwrite, preserving prior profile versions.

Module 2 - Smart Candidate Matching: Evidence-backed candidate ranking against a role, evaluating hard skills, soft skills, past-role relevance, industry background, job level, years of experience, technical competency, education level, certifications, language proficiency, location and nationality. Every dimension carries a configurable weight and a configurable maximum, set per role or role family. Each candidate receives a match score with the score for every dimension, the specific CV passages that support the assessment, and any policy rule (including OIA-approved Omanisation rules) that influenced the result. Nationality is implemented as an explicit configurable policy rule, not as a learned model feature; gender participates in no scoring path.

Module 3 - Assessment Exams: A closed loop from invitation through to graded, comparable results returned to the candidate record and to Jadarah. The platform supports IQ and cognitive tests, multiple-choice, case study, essay and practical items across cognitive, behavioural and discipline-specific competencies for Engineering, Information Technology, Business and Humanities. Delivery runs under a full proctoring stack; objective items are auto-graded, and subjective items receive an AI-proposed score with reasoning and supporting passages for human confirmation. Individual candidate reports and per-position summary reports return to Jadarah, and results integrate with the candidate profile and with interview outcomes.

Module 4 - AI Interviewer: Three closed loops per session: identity confirmation, a two-way interview with dynamic follow-up, and a per-dimension evaluation with evidence and outcome recommendation. HR configures job type, description, duration, preset questions and evaluation dimensions in Jadarah; the platform returns a single-use, time-bounded URL that Jadarah distributes to the candidate. The interview runs in Arabic, English, or a mix within a single session, with a customisable avatar built to OIA's design. Follow-up generation is controlled: for each question, HR-configured dimensions define the scope within which the AI may probe, behind topic guardrails, fully logged. Behavioural indicators (speech rate, pause pattern, volume, fluency, gaze, facial expression, posture, gesture) are reported as observations with evidence links; no non-verbal indicator produces or influences an automated outcome.

Module 5 - Application Scoring and Recommendation: Candidate-facing scoring using the same matching component that serves the recruiter view, with the score derivation shown to the candidate and structured, actionable improvement recommendations constrained to verifiable facts drawn from the profile and the job description. Open roles are reverse-ranked against the candidate's profile with the reason for each match, and all scoring and recommendation activity is logged for audit.



2.3 Benefits and Key Deliverables

- UIS Cloud proposal will provide OIA with the following business benefits and tangible deliverables:

  - **One Integrated Recruitment Capability:** A unified bilingual solution comprising CV Parsing, Candidate Matching, Online Assessment, AI-Powered Interviewing, and Application Scoring and Job Recommendations.

  - **Faster, More Consistent Processing:** Automation of repetitive screening, evaluation, and reporting activities will reduce manual workload and support a more efficient recruitment cycle.

  - **Transparent Decision Support:** Recommendations and scores will be linked to source evidence, approved criteria, and recorded system versions, enabling recruiters to understand each output while retaining final decision authority.

  - **Improved Candidate Experience:** Arabic and English interfaces, accessible interaction options, consistent assessment journeys, and defined fallback channels will support candidates across different roles, abilities, and technical conditions.

  - **Jadarah-Centred Operations:** Secure integration will allow candidate, vacancy, assessment, interview, and result information to move through controlled workflows while Jadarah remains OIA's system of record.

  - **Operational Visibility and Control:** Recruiter dashboards, individual and position-level reports, audit records, performance monitoring, and configurable policies will give authorised teams a consolidated view of recruitment activity.

  - **Production-Ready and Transferable Delivery:** OIA will receive the on-premises solution, complete source code, technical and operational documentation, testing and acceptance evidence, administrator and user training, knowledge transfer, and post-go-live support required for sustained ownership.

2.4 Compliance with RFP Requirements

UIS Cloud confirms full compliance with OIA RFQ 5302/2026, including its mandatory qualification criteria, complete functional scope, implementation obligations, and all four Information Security documents. The proposal has been structured so that each requirement is linked to a defined solution component, delivery activity, acceptance test, or contractual deliverable.

- **Mandatory Requirements:** UIS Cloud provides evidence of comparable AI implementation experience, the required Omanisation documentation, a fully on-premises delivery model, and confirmation of compliance with OIA's Information Security requirements.

- **Functional and Accessibility Coverage:** The proposed solution addresses all five AI modules, Arabic and English operation, responsive access across supported devices, WCAG 2.0 accessibility, administrative controls, dashboards, and reporting.

- **Security and Data Sovereignty:** Candidate information, models, processing, and operational records will remain within the approved Oman-based environment, supported by identity and access controls, encryption, auditability, retention policies, and human decision authority.

- **Jadarah Integration:** Versioned interfaces, controlled field mapping, authentication, acknowledgement, retry, and reconciliation mechanisms will support reliable information exchange while preserving Jadarah as the system of record.

- **Delivery and Acceptance:** UIS will complete the project within the required six-month timeframe, supported by requirements workshops, pilot and user acceptance testing, documented performance and security evidence, training, handover, warranty, and ongoing support.

3 Introduction

This tender submission brings together established Omani participation, international technology delivery expertise, and sovereign AI engineering capability. Modern Arkan ALThiba SPC, a Riyada-registered Omani SME led by Mr. Hamood Al Busaidi, has entered into a joint venture with UIS International LLC (collectively referred to as "UIS"). UIS International brings a long-standing presence in the Sultanate, 46.2% Omanisation, and experience delivering ICT initiatives across government, energy, utilities, and other regulated sectors, environments where availability, auditability, and data protection are core delivery requirements. With more than 250 completed ICT projects in Oman, UIS brings an established local track record. Selected Omani references, including Petroleum Development Oman (PDO) and other major public-sector and utility entities, are listed in Supporting Document 6.1.

The proposed solution will be delivered through UIS Cloud, bringing the specialist cloud-native infrastructure, AI engineering, systems integration, security, and implementation expertise required to translate OIA's requirements into a production-ready capability. Relevant enterprise AI engagements covering AI-assisted recruitment and candidate matching, spoken interview workflows, structured information extraction, human oversight, enterprise integration, operational dashboards, and scalable AI operations are presented in Supporting Document 6.2, "UIS Selected Enterprise AI Project References." Collectively, these engagements were delivered for organisations operating at national and enterprise scale and demonstrate experience under conditions directly relevant to those OIA sets here: regulated or operational data, measurable outcomes, and decisions that must withstand scrutiny after the fact. This experience complements UIS's local accountability with the technical depth and delivery discipline required across design, implementation, integration, testing, adoption, handover, and ongoing support.

UIS's vision is not to implement OIA's five required AI modules as isolated tools. The proposed approach establishes a single, governed recruitment capability integrated with Jadarah, which remains the system of record. The solution will streamline candidate processing, improve consistency across matching, assessment, interviewing, scoring, and recommendation activities, and provide evidence-linked, explainable decision support. Candidate data and AI processing will remain within Oman, while authorised OIA personnel retain final decision authority throughout the recruitment lifecycle.

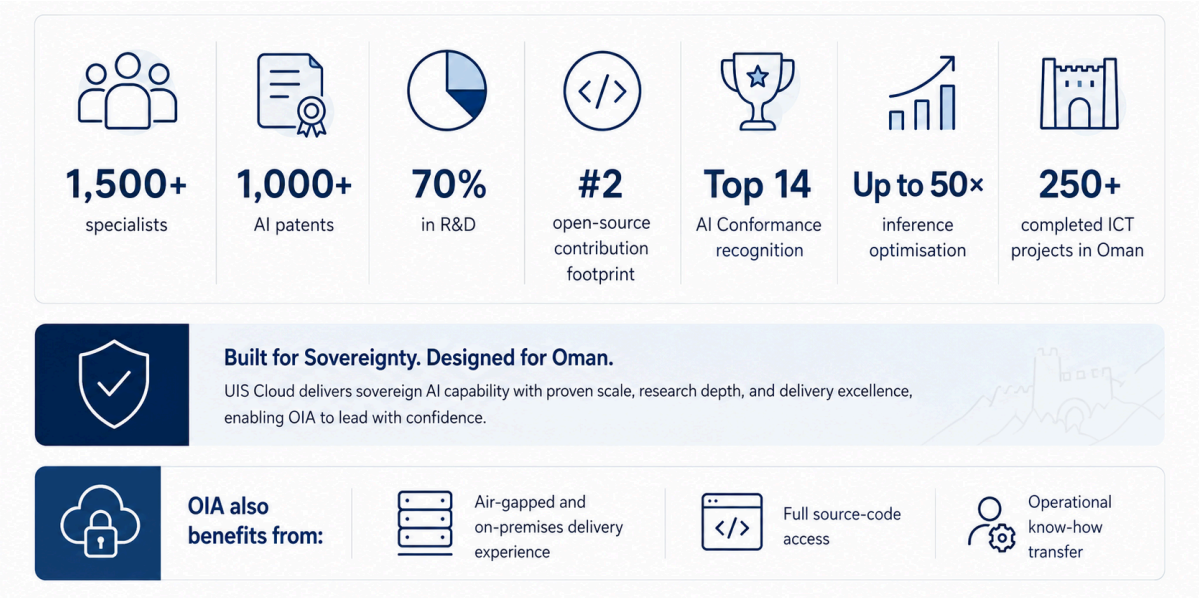
This technical proposal describes the solution architecture, functional coverage, Jadarah integration, delivery methodology, security and AI governance controls, implementation plan, testing and acceptance approach, training, handover, and post-go-live support required to realise that vision. By combining Modern Arkan's local participation, UIS International's established delivery presence in Oman, and UIS Cloud's sovereign AI and implementation capabilities, UIS is positioned to deliver a practical, auditable, and sustainable recruitment solution within OIA's required six-month timeframe, with the operational capability, technical knowledge, and control retained by OIA in Oman beyond project completion.

4 WHY UIS Cloud

Few providers combine applied AI engineering with the platform expertise required to operate it reliably under sovereign conditions. UIS Cloud does both. Behind it is an engineering and AI ecosystem of more than 1,500 specialists across cloud, AI, and platform engineering, supported by over 1,000 AI patents. Seventy percent of this workforce is dedicated to research and development, with AI as its primary focus.

This depth is reinforced by an open-source contribution footprint ranked second only to Google globally. The underlying technology is also recognised among the Linux Foundation's Top 14 AI Conformance platforms and has demonstrated inference optimisation of up to 50× over standard open-source baselines under defined test conditions. For OIA, the value extends beyond scale and speed: the same capability will be applied to translate recruitment policies into controlled workflows, validate Arabic and English models, and manage deployment, versioning, monitoring, and controlled change. Defined AI functions will target accuracy levels reaching 95%, measured against agreed datasets and acceptance criteria.

UIS Cloud also brings proven delivery experience in air-gapped, on-premises environments for governments and financial institutions. Full source-code access and operational know-how transfer will be provided to OIA as standard, enabling its teams to operate, maintain, and extend the solution independently. This combination of applied intelligence, engineering scale, sovereign operation, and long-term ownership is what makes UIS Cloud particularly suited to OIA's recruitment programme.



4.1 TECHNOLOGICAL PROGRESS

UIS Cloud brings together the disciplines required to engineer OIA's recruitment solution across AI models, business rules, real-time services, and operational controls.

- **Purpose-Built AI Engineering:** Each function is assigned to the appropriate mechanism. Deterministic logic governs eligibility, assessment timing, scoring boundaries, and interview state, while specialist AI handles document understanding, semantic matching, speech, and contextual dialogue.

- **Controlled and Traceable Intelligence:** Every scored output can be traced to its source evidence and applicable model, prompt, rule, and question versions. Low-confidence or consequential results are routed for confirmation rather than treated as final conclusions.

- **Operational Resilience:** Dedicated real-time capacity protects live interviews from batch workloads. Defined recovery paths, production-equivalent testing, and release gates ensure that performance, fairness, and continuity are demonstrated before deployment and monitored thereafter.

4.2 POC WILLINGNESS

UIS is prepared to prove the proposed solution, not merely describe it. At OIA's request, we are ready to conduct a focused Proof of Concept using approved anonymised or synthetic data, under jointly agreed scope, timeframe, and success criteria, to demonstrate priority capabilities such as CV parsing and matching, evidence-linked scoring, and Jadarah integration through an available sandbox or agreed interface simulation.

4.3 UNWAVERING COMMITMENT TO CLIENT SUCCESS

For UIS, success is not defined by software installation alone. It is achieved when agreed outcomes have been demonstrated, OIA's teams are prepared to operate the capability, and the service remains stable after launch.

- **Outcome-Led Governance:** The programme will operate against an agreed baseline with named owners, decision dates, and escalation routes. Progress, risks, dependencies, and changes will be reported transparently, while acceptance will be based on signed evidence rather than unsupported claims.

- **Capability Transfer:** Role-based training, runbooks, as-built documentation, administration guides, and technical knowledge transfer will prepare OIA's business and IT teams to manage daily operations and controlled future changes.

- **Post-Go-Live Accountability:** UIS will provide the three-month warranty and one-year support period required by the RFQ, supported by local first-line capability, specialist escalation, and continued monitoring. Incidents, defects, and agreed improvement actions will remain tracked through resolution.

4.4 KNOWLEDGE TRANSFER AND OMANI PARTICIPATION

Our measure of success is not only a successful go-live, but OIA's ability to operate and evolve the platform without unnecessary long-term dependency on its supplier. Knowledge transfer is therefore treated as a delivery outcome, not a handover event. OIA teams will work alongside UIS Cloud specialists throughout implementation, while Omani participation is embedded in operational readiness, local support, and structured mentorship across AI operations and integration.

Through hands-on training, technical walkthroughs, and guided operation, OIA will gain the practical capability to manage, maintain, adapt, and independently validate the solution. The handover will include the delivered application source code, configuration, deployment and acceptance assets, and version-controlled operational documentation. OIA will therefore receive more than a working platform: it will retain the knowledge, control, and confidence to shape how that platform evolves, with the capability firmly anchored in Oman.

4.5 AI-POWERED CASE STUDIES AND SUCCESS STORIES

The following three case studies have been selected for their specific relevance to OIA's requirements. One covers a closely comparable recruitment process, while the other two demonstrate supporting AI capabilities at enterprise scale, including transforming unstructured and spoken inputs into controlled outputs, integrating AI with existing systems, and maintaining human verification within operational workflows.

Rather than presenting broad corporate references, each case study identifies the client challenge, the delivered solution, the measurable outcome, and its precise relevance to RFQ 5302/2026. Together, they demonstrate that the proposed approach is grounded in implementation experience, while its final configuration and performance will be validated against OIA's data, policies, and acceptance criteria.

4.5.1 REFERENCE 1: ON-PREMISES CLINICAL SPEECH AND STRUCTURED RECORD WORKFLOW

Challenge: The hospital required clinical conversations from ward rounds, handovers, and consultations to be converted into structured records despite background noise, multiple speakers, regional dialects, and specialised terminology. Confirmed information also had to reach the existing hospital system without duplicate entry.

Solution: UIS Cloud delivered a phase-one, on-premises AI platform across 15 capture terminals. The solution combines streaming speech processing, speaker and role identification, terminology correction, clinical entity extraction, and structured-record generation. Each draft is reviewed by a clinician before confirmed information is written back through controlled field mapping, acknowledgement, and retry mechanisms.

Results: The engagement established a complete and auditable workflow from spoken interaction to filed record. AI-generated drafts cannot enter the system without clinician confirmation, corrections remain traceable, and failed interface transactions are automatically returned and retried. This provides a proven operating pattern for OIA's spoken interview processing, human-controlled validation, and reliable integration with Jadarah.

4.5.2 REFERENCE 2: END-TO-END AI RECRUITMENT ACROSS TWELVE SUBSIDIARIES

End Client: Hunan Feiwo New Energy Technology Co., Ltd.

Challenge: Twelve subsidiaries were recruiting independently, resulting in inconsistent evaluation practices, limited group-level visibility, and different qualifying rules. The client also needed to screen specialist candidates at scale while maintaining data separation and validating performance against live services and client-labelled data.

Solution: UIS Cloud implemented an AI-assisted recruitment programme covering role-requirement decomposition, job-description generation, multi-channel CV aggregation, candidate screening and matching, profile generation, spoken interview assistance and transcription, cross-entity reporting, and recruitment-stage monitoring. The programme comprised seven use cases delivered through twelve independently tested capability modules**.**

Results: Eleven of the twelve modules passed their acceptance gates on live data. CV screening was verified against an agreed 90% accuracy threshold, spoken-interview transcription achieved a word error rate of 5% or less, and the full acceptance criteria for job-description generation and multi-entity governance were met. This is UIS's closest reference to OIA's scope, directly reflecting the workflow covered by the CV Parsing, Matching, AI Interviewer, and Recommendation modules.

4.5.3 REFERENCE 3: HIGH-VOLUME AI CONTENT AUTOMATION AND OPERATIONAL REPORTING

Challenge: Producing visual and written assets manually could not serve a catalogue exceeding 1.3 million items. Outputs also had to meet different requirements across six commercial platforms and remain consistent throughout more than 200 online storefronts.

Solution: UIS Cloud developed a 13-module AI content production and distribution proof of concept. A catalogue-identifier-triggered pipeline generates visual and written assets for multiple platforms, routes work through a scalable batch queue, and provides continuously refreshed operational dashboards. Each processing stage is independently testable and supported by containerised deployment, monitoring, and controlled operator oversight.

Results: The accepted proof of concept sustained production above 3,000 assets per day across all six platforms while retaining review controls and storefront-level reporting. The engagement demonstrates the high-volume processing, workload orchestration, monitoring, and dashboard capabilities relevant to OIA's CV-processing and assessment peaks.

4.5.4 RELEVANCE TO OIA'S AI-DRIVEN RECRUITMENT ASSISTANT

Together, these engagements demonstrate the delivery capabilities most relevant to OIA's programme:

Direct Recruitment Experience: The Hunan Feiwo engagement covers the recruitment chain from role definition and CV screening through candidate matching, spoken interviews, and consolidated reporting.

Human-Controlled AI: The Shanghai Jiabo implementation demonstrates how AI-generated outputs can remain subject to authorised review, with every confirmation and correction retained as auditable evidence.

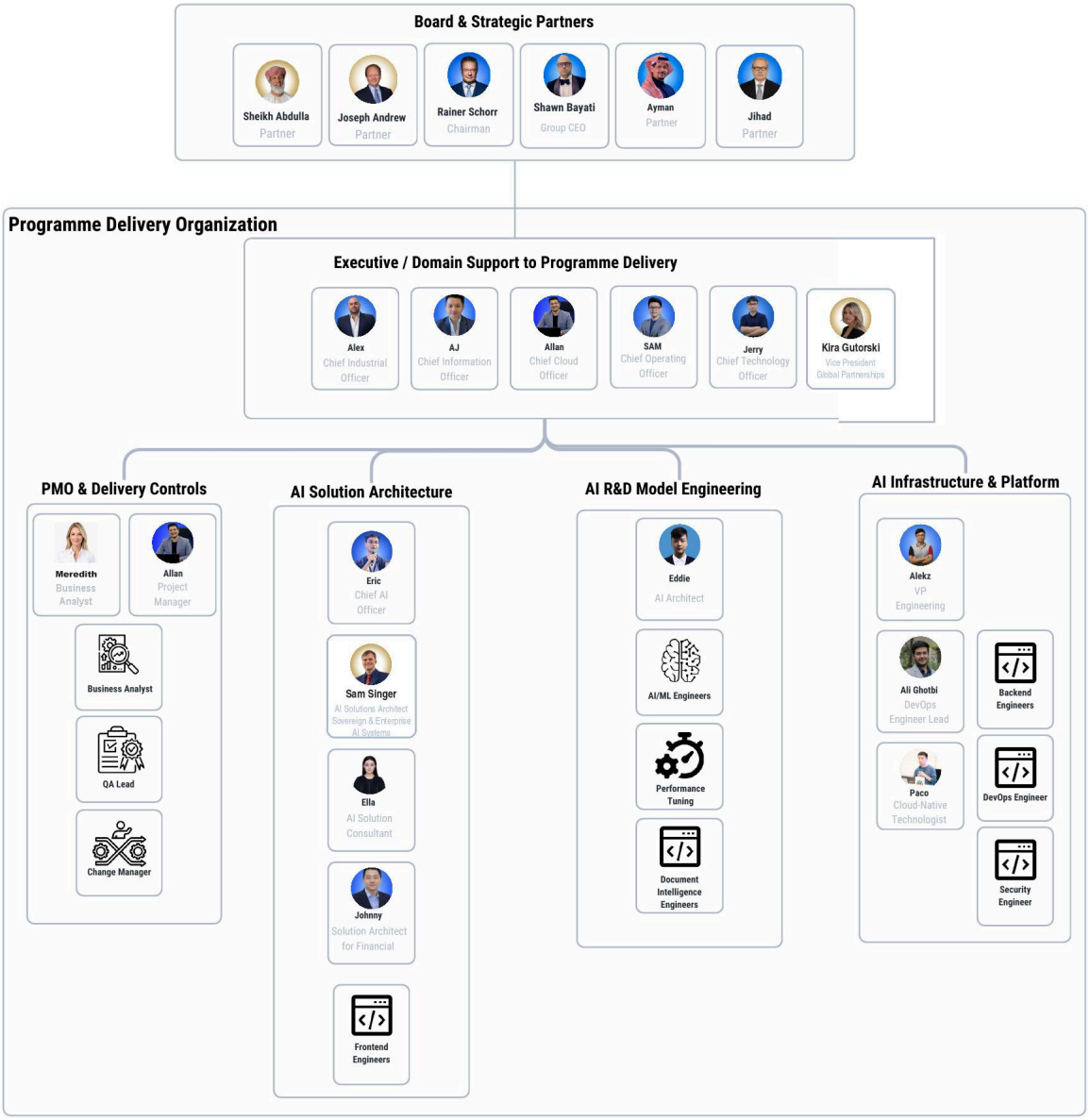
System-of-Record Integration: Proven field mapping, acknowledgement, retry, and reconciliation patterns provide a practical foundation for dependable integration with Jadarah.

Enterprise-Scale Operations: The references collectively demonstrate real-time speech processing, high-volume batch workloads, multi-entity governance, operational dashboards, and acceptance testing against live or client-labelled data.

Operational Ownership: On-premises deployment, structured knowledge transfer, and application-layer handover support OIA's requirement for long-term control of its technology, data, and operating capability.

These references reduce implementation risk by showing that the underlying workflows and engineering patterns have been delivered before, while OIA-specific policies, Arabic and English performance, and acceptance targets will be validated within this engagement.

5 Team Structure



5.1 ROLES AND RESPONSIBILITIES OF TEAM MEMBERS

The following named role profiles, prepared from the supplied CVs, define the proposed team's experience and responsibilities for this engagement.

▌AI/ML Lead and Model Governance Authority — Responsible AI Strategy and Production Assurance

Eric, Chief AI Officer at UIS Cloud, holds a master's degree in Machine Learning and Distributed Computing and has proven experience leading enterprise AI strategy and solution architecture. For this engagement, he will govern model selection and define the criteria for accuracy, fairness, explainability, human oversight and production approval. He will direct validation, monitoring and retraining controls across the modules, drawing on his experience in AI document intelligence, agent orchestration, decision automation and national AI-policy compliance.

▌Business Strategy and Benefits Realisation Lead — KPI Alignment and Executive Value Governance

SAM, Vice President of Strategy Management at UIS Cloud, has proven experience in corporate strategy, market intelligence and financial-sector technology positioning. For this engagement, SAM will translate [OIA] objectives into measurable KPIs, align business and scoring rules with process owners, and maintain a clear link between delivered capability and expected business benefit. The role will also govern benefits-realisation reviews and executive reporting, supported by experience in AI, cloud and financial-services strategy and PMP certification.

▌Executive Technology and Data Governance Lead — Sovereign Strategy and Responsible Oversight

AJ Mak, Chief Information Officer at UIS Cloud, brings 10 years of AI expertise and 20 years of proven international business-expansion experience across more than 25 countries. For this engagement, he will align the solution with [OIA]'s technology roadmap and provide executive oversight of data ownership, access, human oversight and cybersecurity alignment. He will also review capability transfer and long-term platform direction, drawing on his experience in AI implementation, predictive analytics, cloud strategy and digital transformation.

▌Technical Director and Architecture Authority — Sovereign Architecture and Technical Governance

Jerry**,** Chief Technology Officer at UIS Cloud, brings more than 15 years of proven experience in cloud-native architecture, cybersecurity, AI and application modernisation. For this engagement, he will approve the end-to-end on-premises architecture and govern integration with [The SYSTEMS], APIs and SSO. He will provide final technical authority for security, high availability, disaster recovery, auditability and acceptance decisions, ensuring that the proposed solution remains scalable, compliant and operable within [OIA]'s controlled environment.

▌Lead Solution Architect — Requirements Translation and Enterprise Design

Kenneth brings 21 years of proven experience in enterprise AI platform architecture, intelligent document parsing, knowledge systems, question generation and rule-checking applications. For this engagement, he will translate workshop findings and requirements into the end-to-end solution design, defining functional boundaries, data flows and dependencies across all [5] modules. He will maintain architecture documentation and technical traceability from requirement through implementation and acceptance, while coordinating the interfaces between services, enterprise systems and operational workflows.

▌AI Interaction and Evaluation Architect — Auditable Agentic Workflows and Evaluation Controls

Sam Singer is an AI Solutions Architect with more than five years of proven production-engineering experience across agentic AI, retrieval-augmented systems and regulated enterprise delivery. For this engagement, he will design the adaptive interaction flow, follow-up logic, auditable summaries and scoring controls. He will also establish automated evaluation gates, grounded-output checks, self-hosted model-access controls and complete request logging so that generated outputs can be reviewed and traced within a sovereign environment.

▌Document Intelligence and Integration Solution Architect — Extraction and Secure Workflow Integration

Johnny is a Solution Architect with 12 years of proven experience in enterprise software and AI-based financial document extraction, processing and dashboard solutions. For this engagement, he will design ingestion, OCR and structured extraction, map source data to agreed target fields, and define confidence thresholds, manual-review paths and exception handling. He will also govern interfaces and data exchange with [OIA SYSTEMS], ensuring that extracted information remains traceable from its source document to the approved record.

▌Analytics and Responsible AI Specialist — Human-Centred Measurement and Fair Evaluation

Ella is an AI Solution Consultant and quantitative analyst with graduate training in Quantitative Analytics and an academic background in psychology. Her proven experience covers behavioural research, statistical modelling, survey design and multilingual data validation. For this engagement, she will structure scoring and reporting, apply quantitative methods to evaluate reliability and impact risk, and define fairness checks, human-review controls and acceptance criteria so that outputs remain interpretable, appropriately governed and defensible.

▌Document AI and Model Validation Engineer — Pipeline Engineering and Accuracy Assurance

Eddie is an AI Architect with a master's degree in Electronic Information and proven experience developing enterprise document-processing pipelines. For this engagement, Eddie will build the preprocessing, OCR and extraction workflow, including dataset preparation, train–validation–test separation and model integration. The role will also cover K-fold cross-validation, confusion-matrix analysis, drift detection, retraining triggers, model cards and validation reports, supporting measurable performance, disciplined UAT and a documented handover.

▌Platform Engineering, Data and Storage Lead — Resilient Cloud-Native Data Infrastructure

Alekz, Vice President of Engineering at UIS Cloud, brings more than 14 years of proven experience in cloud-native platforms, big-data infrastructure and software-defined storage. For this engagement, he will design the secure on-premises data and storage services supporting the solution, including replication, backup, recovery and high-availability controls. He will also optimise vector and document-data pipelines and lead capacity, performance and resilience testing, drawing on his leadership of the Kubernetes-native HwameiStor and Piraeus storage projects.

▌Cloud-Native and Kubernetes SME — Upstream Leadership and Workload Reliability

Paco is an open-source and cloud-native technologist with more than 10 years of proven experience in Kubernetes, container orchestration and AI infrastructure. For this engagement, he will review cluster topology and lifecycle management and advise on workload scheduling, scaling, multi-tenancy and resource governance. He will support platform hardening, complex troubleshooting and knowledge transfer to [OIA]'s technical team, drawing on his elected leadership and maintainer roles across the Kubernetes and CNCF communities.

▌DevOps, CI/CD and Platform Operations Lead — Secure Automation and Operational Readiness

Ali Ghotbizadeh is a DevOps Engineer Lead with five years of proven experience in Kubernetes, CI/CD, infrastructure automation and observability. For this engagement, he will implement delivery pipelines, Infrastructure as Code and GitOps practices across clearly separated development, testing and production environments. He will configure secrets management, monitoring and logging, release and rollback controls, backup procedures and operational handover. His CKA and CKS credentials and experience sustaining high-availability Kubernetes platforms support secure, repeatable and auditable deployment.

▌Project Manager and On-Premises Cloud Lead — Integrated Delivery and Infrastructure Readiness

Allan Salmaji, Chief Cloud Officer at UIS Cloud, brings more than seven years of proven experience designing and operating cloud and on-premises platforms. For this engagement, he is proposed to manage the integrated delivery plan, RAID log and work packages across the modules, coordinating infrastructure readiness, testing, training and handover. He will govern reporting, dependencies and escalation while connecting project decisions to the practical requirements of Kubernetes, automation, high availability and disaster recovery throughout the delivery lifecycle.

▌Commercial and Stakeholder Governance Lead — Institutional Partnerships and Bid Assurance

Kira Maxima Gutorski, Vice President of Global Partnerships at UIS Cloud, has proven experience leading institutional relationships from qualification and structuring through signature and joint delivery. For this engagement, she will coordinate stakeholder engagement, workshops and decision logs and govern commercial compliance, risk evidence and joint-venture commitments. She will also maintain the verification chain for client-facing material, ensuring that technical and commercial statements are source-checked and internally approved before submission or formal decision.

In Country Value

As a testament to our dedication to Oman's growth and development, we are eager to contribute to In-Country Value (ICV) initiatives by providing valuable training and mentorship opportunities to Omani nationals. This commitment to fostering local talent and expertise aligns with our broader goal of supporting a sustainable and prosperous future for the Sultanate.

6 Pre-Implementation Phase

The Pre-Implementation Phase will establish a clear, jointly agreed foundation for delivery. During Weeks 1–2, UIS and OIA will align key stakeholders, confirm access and security prerequisites, validate the initial solution configuration, and verify the assumptions affecting performance, integration, and infrastructure sizing. By the end of Week 4, both parties will formally approve the requirements, Jadarah integration scope, acceptance criteria, dependencies, and project baseline. This early alignment will reduce delivery risk and enable implementation to proceed with clear responsibilities and measurable outcomes within the required six-month programme.

6.1 Stakeholder Analysis and Engagement

- **Objective:**

Establish a shared understanding of OIA's current recruitment model and ensure that business priorities, technical constraints, governance requirements, and user needs are reflected in the solution from the outset.

UIS will conduct structured workshops and working sessions with the relevant stakeholder groups:

- **HR and Recruitment:** To define the end-to-end recruitment process, screening and assessment practices, interview workflows, decision authority, and reporting needs.
- **IT:** To confirm Jadarah interfaces, identity and access requirements, infrastructure boundaries, security standards, and deployment procedures.
- **Admin and Business Support:** To understand administrative workflows, document handling, and record-keeping obligations.
- **Compliance and Data Protection:** To establish requirements for data governance, retention, masking, auditability, and regulatory alignment.
- **Candidate Perspective:** To ensure that candidate journeys, communication, accessibility, and Arabic and English interactions are appropriately designed.

The findings will be translated into documented workflows, user journeys, requirements, use cases, and acceptance criteria. Each requirement will remain traceable to its stakeholder or RFQ source and to the test evidence used to verify its delivery.

6.2 Technical Requirements Gathering and Current System Assessment

Objective:

Establish a verified understanding of OIA's existing recruitment environment and define the functional and technical baseline for the proposed solution.

Activities:

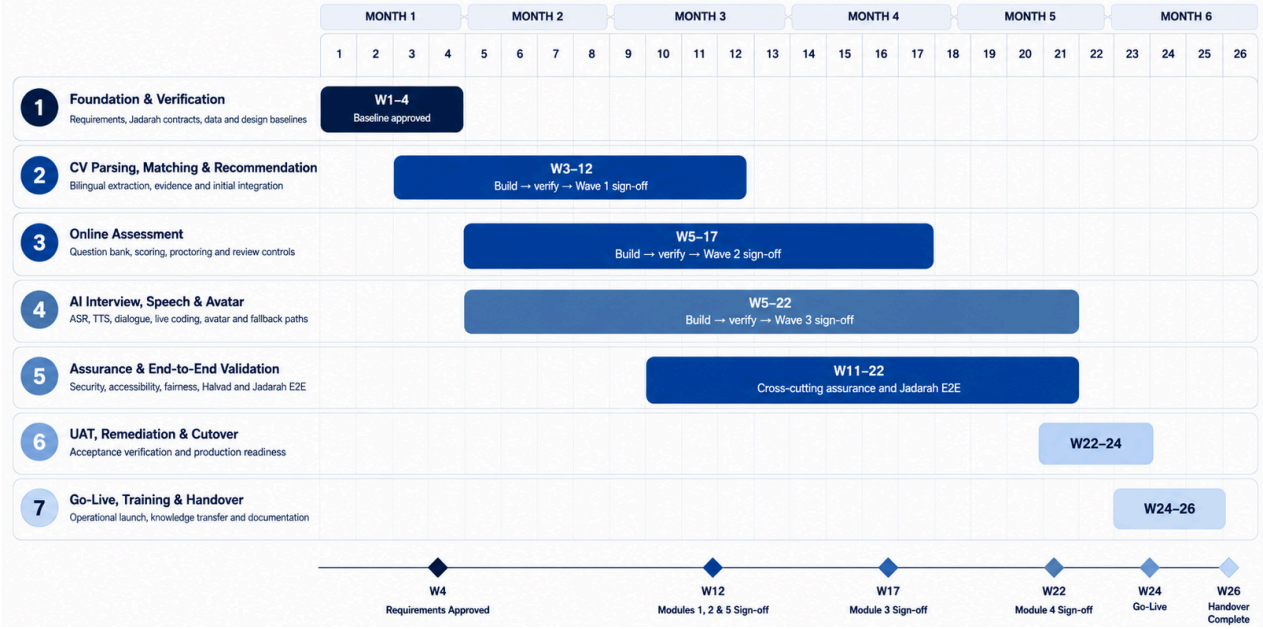
- **Current Environment Assessment:** Review Jadarah, existing recruitment workflows, available data, interfaces, identity and access arrangements, hosting environment, network boundaries, security controls, and deployment procedures. This assessment will identify the constraints, dependencies, and gaps affecting implementation and integration.

- **Requirements Definition:** Document the functional requirements for the five AI modules and their interaction with Jadarah, together with non-functional requirements covering performance, availability, scalability, accessibility, security, privacy, data residency, and auditability. Each requirement will have a defined source, priority, and acceptance criterion.

- **Security and Compliance Alignment:** Confirm that the proposed design addresses OIA's information-security requirements, data-governance and privacy obligations, applicable MTCIT policies, and WCAG 2.0 accessibility requirements.

- **Gap and Dependency Assessment:** Compare the current environment with the proposed target state and document any required inputs, decisions, remediation actions, and ownership. The findings will form the basis for the agreed requirements specification, system landscape, gap analysis, and acceptance baseline.

6.3 Roadmap Development



Objective:

Provide OIA with a controlled six-month delivery roadmap that aligns workstreams, milestone approvals, responsibilities, and critical dependencies from project mobilisation through operational handover.

Activities:

- **Delivery Sequencing:** Organise the programme into coordinated, overlapping workstreams covering foundation and verification, implementation of the five AI modules, Jadarah integration, assurance, user acceptance testing, go-live, training, and handover.

- **Milestone and Acceptance Control:** Link every phase to defined outputs and approval points. The roadmap progresses from requirements approval in Week 4 through module sign-offs in Weeks 12, 17, and 22, followed by solution go-live in Week 24 and completion of handover in Week 26.

- **Accountability and Resource Alignment:** Assign named leads and clear OIA–UIS responsibilities to each workstream, ensuring that business decisions, technical inputs, reviews, and approvals are incorporated into the delivery schedule.
- **Dependency and Risk Management:** Track required inputs and dependencies with an owner, required date, impact, mitigation, and escalation route. Progress, risks, and changes will be reviewed regularly against the approved milestones and acceptance evidence.

6.4 Technical Design and Architecture Planning

Objective:

Define the production architecture, data flows, and engineering controls that will govern the implementation and operation of OIA's five AI recruitment modules.

Activities:

- **Modular and Layered Architecture:** Establish clear boundaries across candidate and recruiter interfaces, recruitment workflows, the five AI modules, Jadarah integration, AI processing services, data and evidence storage, and platform operations. Real-time interview services will remain isolated from batch workloads such as CV processing and offline assessment activities.
- **Jadarah Integration and Data Design:** Define versioned contracts for the five required interface groups and establish a canonical Candidate, Job, Application, and Assessment data model. Idempotency, retry, outbox, and reconciliation controls will protect data exchange from duplication, interruption, and silent loss.
- **Performance, Capacity, and Resilience Planning:** Validate CV throughput, interview concurrency, latency, and infrastructure demand through production-equivalent testing. Application, integration, and data services will be designed for high availability, while live AI services will use an N+1 configuration with active-session recovery.
- **Security, Privacy, and Operational Control:** Incorporate OIA-controlled identity, keys, secrets, and certificates; encryption at rest and in transit; network segmentation; controlled egress; append-only audit evidence; monitoring; backup and recovery; and governed model release and rollback procedures.

The resulting architecture decisions, interface contracts, data model, capacity assumptions, security controls, and acceptance methods will provide the controlled technical blueprint for development, integration, testing, and operational handover.

7 Scope of Work & Deliverables

7.1 General Requirements (5%)

UIS Cloud will deliver a single recruitment platform covering CV parsing, matching, assessment, AI-assisted interviews and job recommendations. The five functions share candidate and vacancy data, assessment criteria, audit records, approval controls and Jadarah integration.

The design supports on-premise, offline operation and the six-month delivery schedule. Items requiring early agreement include Jadarah access, representative Arabic and Omani data, question-bank distribution, avatar and voice approval, the security test window and final hardware configuration. Each item has an owner, a required decision date and a defined escalation route.

- **Mutual NDA signed by all parties including subcontractors** — Accepted. A mutual NDA will be executed by the bidder and by every party contributing to delivery, in accordance with OIA policy and procedure, before any OIA data is accessed
- **All observations from OIA security assessments closed** — Accepted. Assessment windows are scheduled before each release wave at section 7.7; findings are remediated and re-verified before the release proceeds

Technical response commitments:

- Deliver the proposed on-premise, offline and human-governed architecture.
- Complete an embedded Weeks 1-2 configuration and verification sprint to confirm project measurements and the final GPU bill of materials without extending the six-month programme.
- Test with representative Arabic, English and Gulf/Omani datasets using blind, reproducible methods.
- Define service levels at the end-to-end user journey, not at individual model calls.
- Provide documented commercial, on-premise and offline usage rights for every delivered component, with exact licence obligations recorded in the software and asset registers.
- Provide explainable, human-governed multimodal behavioural assessment where required, including speech delivery, gaze, facial expression, posture, gesture, composure, assertiveness, aggression/calmness indicators and response to challenge. No single biometric or behavioural signal will determine a hiring outcome.

Engineering principles:

- **Differentiating capabilities:Jadarah remains the system of record.** The platform retains only the snapshots and execution evidence required to operate and audit the recruitment workflow.
- **Real-time and batch workloads are isolated.** CV OCR and offline analytics cannot consume the GPU capacity reserved for live interviews.
- **Rules and generation are separated.** The LLM may draft a follow-up question or an explanation; versioned rules control hard gates, weights, maximum scores and workflow transitions.
- **Automated results remain traceable.** Every extracted field, match, score and human override links to source content and the applicable model, prompt, rule and question versions.
- **Low-confidence results require confirmation.** The system requests confirmation, repetition, an alternative channel or human review; uncertainty does not penalise the candidate.
- **Fallbacks preserve the interview.** The avatar may fall back to a static image and audio, then to a text channel, without changing the approved interview and scoring logic.
- **Offline operation is designed in.** Production does not depend on public inference APIs, public STUN servers, external telemetry, online activation or dynamic model downloads.

- **OIA retains decision authority.** The platform provides decision support. Final shortlisting, rejection and exception decisions remain with authorised OIA roles.

- **Complete sovereign operation** — Every component — language models, speech recognition and synthesis, avatar rendering, behavioural processing — runs inside Oman on OIA-controlled infrastructure, with no external call and no online licence validation. The environment operates correctly with outbound network access closed, and this is demonstrated as an acceptance test rather than asserted
- **Evidence on every output** — Each score resolves to the passage, segment or answer that produced it. A recruiter can open any figure and see its basis; a candidate who questions a result can be answered from the record
- **Controlled dialogue generation** — Interview follow-ups are generated within dimensions the recruiter has defined, behind topic guardrails, fully logged. This delivers genuinely dynamic conversation while making it structurally impossible for the line of questioning to differ systematically between candidate groups
- **Governance delivered as working tooling** — The fairness test harness, model documentation, decision audit trail and periodic validation reporting are handed over as capabilities OIA continues to operate — not as a compliance annex written once at handover
- **A knowledge model OIA owns** — The business data model, rules, scoring dimensions, weights and question banks are delivered as OIA's asset, editable without reference to the supplier and portable beyond the platform
- **Bounded behaviour where certainty is not available** — Where a model can be wrong — speech recognition, identity matching, observation of non-verbal attributes — the commitment is paired with a defined fallback so an error becomes a review step rather than a wrong outcome for a candidate

7.1.1 The Solution and Why It Is Built This Way

OIA is acquiring an AI layer over an existing recruitment system, not a replacement for it. Jadarah remains the system of record; the solution adds five AI capabilities to it and returns results to it. That framing drives the two decisions that shape everything else in this proposal.

One foundation rather than five applications. The five models are built on a shared business model — one candidate record, one job model, one evaluation vocabulary, one evidence layer. Built separately they would produce divergent views of the same candidate, make explainability impractical, and make each future change five changes. Built on one foundation they produce consistent outputs across the pipeline, allow a score to cite the same underlying facts a recruiter can inspect, and make a sixth capability an addition rather than a new build. This is also what makes six-month delivery credible: the models are not five sequential projects.

Every AI output ends at a person. The RFQ requires that final hiring decisions always be made by a human. Rather than treating that as a constraint applied at the end, the solution is designed so that each use case terminates at a confirmation step, and the record of that confirmation — who, when, on what evidence — is what the system stores. This is what makes the outputs defensible to a candidate who questions them and to a regulator who reviews them.

7.2 System Requirements

This section responds to the System Requirements at RFQ section 4.2.

7.2.1 **Compatible with Jadarah** — Integration design at section 3.7; the platform consumes Jadarah as the system of record and returns results to it, maintaining no parallel master copy

7.2.2 **24/7 availability with high availability and automatic failover** — Application, integration and data services are deployed redundantly with automatic failover. Real-time interview capacity is provided by two independent GPU inference nodes in an N+1 configuration, with pre-warmed models, redundant media routing and replicated control services. Each interview event is journalled with the transcript, question, scoring state, media time point and a secure reconnection token. The client automatically renegotiates media and the orchestrator reconstructs the active context on the surviving node.
An interview in progress will survive an application, media, GPU-process or GPU-node failure without candidate restart or loss of a confirmed turn. A brief media pause may occur while traffic is rerouted and volatile inference context is reconstructed; automatic reconnection resumes from the correct point and preserves the approved question and scoring state. Where the strictest continuity tier is selected, shadow speech processing and hot-standby context replication reduce the interruption further. Gate G14 covers fault injection, continuity, no confirmed-turn loss and the agreed service-recovery objective; the initial target is resumption within 30 seconds, with database and environment RPO/RTO agreed with OIA.

7.2.3 **WCAG 2.0 compliance on all AI interfaces per MTCIT policy** — Conformance across candidate-facing and internal interfaces, with live captioning and an equivalent text interview channel so that a spoken interview is not the only available route. Accessibility test evidence delivered (section 2.6)

7.2.4 **Clear reasons for decisions; staff understand how and why** — Every score resolves to the passage, segment or answer that produced it. The recruiter interface opens the evidence directly from the figure rather than presenting a separate explanation (sections 2.2, 2.4, 2.6)

7.2.5 **Final hiring decisions are always made by a human** — Every use case terminates at a confirmation step; no model issues an outcome. Shortlisting requires a written reason; interview outcome recommendations are presented for a recruiter to accept, adjust or reject; identity and proctoring detections are flags for adjudication, never automatic disqualification (section 2.6)

7.2.6 **Trained on diverse and representative data; regularly tested for freedom from unfair bias** — Calibration uses OIA's own historical data, which is representative of the population the system will serve. Symmetric-pair testing across gender and nationality is built into the delivered solution. Nationality operates as an explicit, configurable, auditable policy rule rather than a learned feature; gender participates in no scoring path (sections 2.2, 2.6)

7.2.7 **Routine review and validation ensuring equal performance across user groups over time** — Fairness revalidation on the agreed cycle through the support period, reported as a named deliverable, with the test harness operated by OIA after handover (section 10.3)

7.2.8 **Logs of decisions maintained for traceability and auditing** — Inputs, applied rules, AI recommendation and human confirmation recorded for every decision in an append-only log with integrity verification, reconstructable on request and exportable for regulatory review (section 5.3)

7.2.9 **Continuous performance monitoring including drift detection, accuracy and precision tracking** — Monitoring operates continuously with alerting; drift, accuracy and precision are tracked per model and reviewed quarterly, with recalibration where indicated (section 10.3)

7.2.10 **Installation requirements specified — hardware, software, infrastructure** — Architecture at section 5.2; reference hardware configuration, software stack and infrastructure prerequisites at Annex C

7.2.11 **All required licences covered by the bidder** — Accepted. All licences necessary to operate the solution are provided by the bidder and are set out at Annex B

7.2.12 **Licences activated post production deployment** — Accepted. Activation follows production deployment and is performed offline, with no dependency on an external validation service (sections 5.4, Annex B)

7.2.13 **Role-based access control restricting admin and internal user functions** — Section 4.2, with least privilege, privileged-action logging and masked views by default for operational roles

7.2.14 **Build and release management — phased deployment and rollback, standardised DevOps with build numbering and pipeline management, monitoring and logging framework** — Section 5.6

7.2.15 **Infrastructure scalable across interfaces, data volumes and additional use cases; horizontal scaling under increasing load** — Section 9.3

7.3 Functional Requirements (35%)

The RFQ specifies five AI models. They are not built as five separate applications. They share one candidate model, one job model and one evaluation vocabulary: parsing produces the structured facts that matching and scoring both consume; assessment and interview results return to the same candidate record; the scoring model is the candidate-facing view of the same matching component the recruiter sees in the recruiter view.

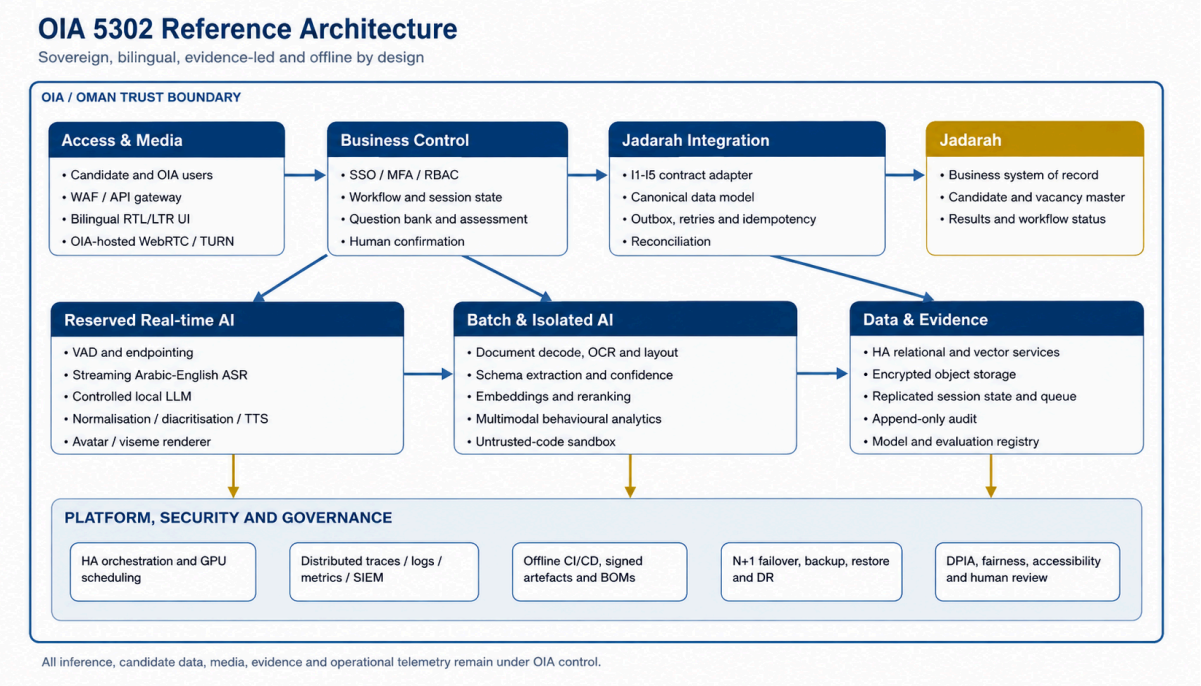


Figure 1. *All candidate data, media processing, inference, audit records, model artefacts and operational telemetry remain within the OIA-controlled Oman boundary. Jadarah is integrated through a contract-first adapter and remains the business system of record.*

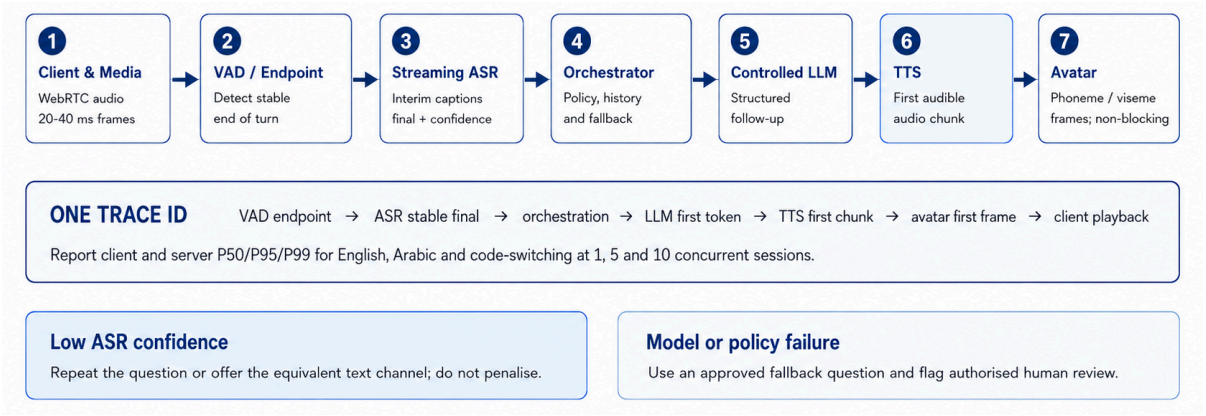


Figure 2. *Latency is measured from the candidate's end-of-speech event to the first audible response on the same client. A single trace ID links the complete chain, including fallback and interruption events.*

This matters functionally, not just architecturally. It is what makes a match score, an assessment result and an interview evaluation refer to the same candidate facts rather than three divergent copies. It is also what makes explainability tractable — a score can only cite evidence if every module refers to the same underlying record.

Shared elements and the models that consume them:

- **Candidate record — profile, CV versions, application history** — All five models
- **Job model — description, level, requirements, evaluation dimensions and weights** — Matching, assessment, interview, scoring
- **Relevance engine — dimensional scoring with evidence location** — Matching and scoring
- **Evidence and explainability layer** — All scored outputs
- **Human confirmation and audit layer** — Every decision point

Business capability scope:

Business model	Primary outcome	Core technical capabilities
Model 1 - CV Parsing	A verified, structured candidate profile from new and historical CVs	Native text extraction, Arabic/English OCR, layout analysis, structured fields, confidence, evidence spans, CV comparison and candidate confirmation
Model 2 - CV-to-Job Matching	Explainable candidate-to-vacancy suitability	Bilingual taxonomy, hybrid retrieval, versioned rules and weights, evidence-backed scoring, gap analysis and recruiter confirmation
Model 3 - Online Assessment	Secure, bilingual and job-relevant assessment	Governed question bank, blueprints, deterministic and rubric-based scoring, coding sandbox, identity/proctoring flags, accessibility and psychometric review
Model 4 - AI Interviewer	Structured bilingual interviews with controlled follow-up	WebRTC media, streaming ASR, controlled dialogue, TTS, avatar, captions, interruption handling, evidence-based scoring and recovery
Model 5 - Job Recommendation	Suitable vacancy recommendations for candidates	Reuse of the same matching engine, job eligibility rules, evidence, explanations and candidate-facing guidance

Functional behaviours common to all five models:

- **Humans make the final decision** — Every AI output terminates at a human confirmation step. No model issues a hiring outcome. Recommendations and rankings are presented for a person to accept, adjust or reject, and the person's action is what is recorded
- **Explainability** — Every score carries verbatim evidence resolving to its source — a passage in a CV, a segment of a transcript, an answer in an assessment. A recruiter can open any figure and see what produced it
- **Bilingual operation** — Interfaces, question banks, reports, notifications and training materials are delivered in Arabic and English, with the architecture prepared for further languages
- **Accessibility** — WCAG 2.0 conformance across candidate-facing and internal interfaces; live captioning and an equivalent text interview channel so that a spoken interview is not the only route available
- **Fairness** — Symmetric-pair testing across gender and nationality is built into the delivered solution and run periodically, with reports as a named deliverable. Nationality operates as an explicit policy rule; gender participates in no scoring path
- **Audit** — Inputs, applied rules, AI recommendation and human confirmation are recorded for every decision, reconstructable on request and exportable for regulatory review
- **Monitoring** — Continuous performance monitoring including drift detection and accuracy and precision tracking, with periodic revalidation through the support period
- **Reporting** — Tailored reports based on assessment results, with an interactive dashboard covering overall candidate and position detail

Jadarah integration:

Integration model. Jadarah remains the system of record for candidates, jobs and recruitment status. The platform is a consumer of that record and a producer of AI-generated results returned to it. No parallel master copy of candidate or job data is maintained; everything held is derived, reproducible and traceable to Jadarah as its source.

Interface set. Five interfaces constitute the integration scope.

- **Interface 1 - Candidate profile read**: Jadarah → platform — On CV submission and on demand — Candidate identifier, existing profile fields — Retry with backoff; on persistent failure the candidate is queued and the recruiter notified
- **Interface 2 - Candidate profile write-back**: Platform → Jadarah — After the candidate confirms the parsed profile — Structured fields with change set — Idempotent by profile version; unconfirmed writes are replayed at the next reconciliation
- **Interface 3 - Job data read**: Jadarah → platform — When a role enters screening, and on job update — Job identifier, description, level, requirements — Cached within a defined freshness window; stale reads are flagged rather than used silently
- **Interface 4 - Interview creation and unique URL return**: Bidirectional — HR creates an interview in Jadarah — Inbound: job, duration, preset questions, evaluation dimensions. Outbound: single-use, time-bounded session URL — URL generation is synchronous; on failure the creation is rejected with an explicit error rather than left pending
- **Interface 5 - Result return**: Platform → Jadarah — On completion of matching, assessment or interview — Scores, evidence references, report location, status — Queued and retried; every result carries a correlation identifier for reconciliation

Authentication and token flow. Recruiter and administrator access runs through OIA's identity provider using single sign-on with multi-factor authentication, with role-based access control governing administrative and internal functions on the platform side. Candidate identity is issued and managed by Jadarah — the platform receives and validates a Jadarah-issued candidate token and never creates a separate candidate credential. Service-to-service calls use mutual authentication over TLS with tokens issued in accordance with the OIA API Security Requirements.

Unique URL sequence. HR configures the interview in Jadarah → Jadarah calls I4 with the job and interview parameters → the platform creates the session and returns a single-use, time-bounded URL → Jadarah distributes the URL to the candidate → the candidate opens the URL, presents the Jadarah-issued token and completes identity verification → results return through I5.

Reconciliation. A scheduled reconciliation compares interview, assessment and result status on both sides and reports discrepancies for resolution. This covers the case in which a callback is lost and neither side registers an error.

The integration is organised around five interface groups: candidate read, candidate update, vacancy read, assessment/interview creation and results return. The adapter uses versioned OpenAPI/AsyncAPI contracts, idempotency keys, retry with dead-letter handling, an outbox pattern and scheduled reconciliation. A canonical Candidate/Job/Application/Assessment data model prevents point-to-point field drift.

Integration completion depends on early access to the Jadarah API specification, authentication pattern, data dictionary, sandbox, test identities and error/rate-limit behaviour. The contract and test pack will be frozen by the end of Week 2.



7.3.1 Parse CV

What it delivers. Two closed loops: a structured candidate profile created from an uploaded CV and confirmed by the candidate; and a differential update when a candidate uploads a revised CV.

Extraction. The parser accepts PDF, Word and text documents in any layout, including scanned and image-based files, and extracts personal details, educational background, work experience, skills and certifications. Document language is detected automatically and language-specific parsing rules applied — Arabic documents are handled natively, including right-to-left layout, Hijri and Gregorian dates appearing in the same document, and variant Latin transliterations of Arabic names. Extracted values are mapped to standardised internal fields and to the corresponding Jadarah profile fields.

Validation and candidate confirmation. Every extracted field carries a confidence value. The candidate is presented with a step-by-step interface showing three visual states — successfully parsed, requiring review, and missing — and may edit any value or supply a value the parser did not find. Progress is saved as a draft so the candidate may return later. A final summary screen precedes a "Review and Submit" action, after which the confirmed profile is written back to Jadarah. No parsed value reaches the candidate record without the candidate having seen it.

Re-upload. When a candidate uploads a new CV, the system compares the extraction against the existing profile field by field and presents only what has changed, with the old and new values side by side. The candidate decides which changes to accept. Fields not confirmed retain their previous values, and prior profile versions are retained.

The pipeline uses native PDF/DOCX text where available, with page rendering and Arabic/English OCR as the fallback. Layout and reading order are retained before schema-constrained extraction and deterministic validation. Each accepted field carries its source page, text span or image region, original value, normalised value, confidence and model/rule version. Critical or low-confidence fields are confirmed before Jadarah is updated.

The validation set must separate Arabic from English and electronic documents from scans. This is essential because a single blended accuracy value can conceal the conditions most likely to require human intervention.

Acceptance. Field-level extraction accuracy is measured by manual sampling against an agreed field list, reported separately for English and Arabic, and separately again for machine-generated versus scanned documents. Ongoing accuracy is monitored through the support period and improved against the corrections candidates make.

The five implementation activities below are described once, under this first model. The Base document states that requirements gathering and design is conducted once across all five models rather than repeated per model; the same applies to the development, integration, deployment and post-implementation activities, which are run as one programme. Model-specific content is placed under each model's own activity headings.

7.3.1.1 Requirement gathering and design phase

Conducted once across all five models rather than repeated per model, because the models share a foundation. Workshops with IT, HR, Admin and compliance capture data requirements, access needs and security standards. Data transformation requirements, data mapping, validation rules and integration points are defined for every connected system. The integration architecture is documented with all components, interfaces and data flows, designed for scalability and flexibility and aligned to OIA's existing IT estate and long-term direction. Integration with Jadarah is specified in this activity and confirmed against the actual interface documentation.

Test data: At least 300 Arabic and 300 English CVs, stratified by electronic/scanned source and including bilingual, two-column, tabular, low-resolution, rotated, Hijri/Gregorian and mixed-script cases. Ground truth will use two annotators plus adjudication.

Metrics: OCR CER/WER, field precision/recall/F1, critical-field exact match, document success rate, confidence calibration and human-review rate, all reported separately by language and document class.

Targets: Critical-field exact match of at least 95% English and 92% Arabic for electronic documents, and at least 90% English and 85% Arabic for scans. The signed acceptance plan will freeze field definitions, document classes and confidence intervals.

Review path: Weeks 1-2 establish separate Arabic and English results and confirm the production configuration. Results are not blended across languages or document classes. Native document text is preferred; OCR is used where required; low-confidence and critical fields are shown with their source for candidate or authorised-user confirmation.

7.3.1.2 Application Development and testing

Each model is developed against the signed requirements baseline, with interfaces for both recruiters and candidates. Development proceeds in the wave structure above, with continuous integration, automated testing against the acceptance criteria, and internal quality gates before any build is proposed for a release.

- **Model 1** — Core parsing capable of reading, interpreting and extracting structured data from CVs in various formats, minimising manual entry; candidate review interface allowing review, correction, addition of missing detail, and final submission; recruiter-facing interface

Implementation: The modular pipeline uses native text first, followed by Arabic/English OCR, layout reconstruction, schema-constrained extraction, deterministic validation, calibrated confidence and candidate confirmation. PaddleOCR-VL-1.6 is the reference for layout-aware document parsing and complex page elements; PP-OCRv5 Arabic provides a deterministic Arabic text-recognition path. A locally deployable substitute may be used only if it matches or improves the agreed accuracy results and supports offline operation with suitable usage rights.

Throughput will be measured for the complete extraction chain, not OCR alone. A 60-minute production-equivalent test will report pages/minute, document latency, error/retry rate, queue depth, CPU/GPU/memory utilisation and scaling efficiency at 1/2/4/8 workers.

UIS Cloud will establish the complete-chain rate on the selected production-equivalent hardware during Weeks 1-2 and repeat the test with one worker node unavailable. Additional stateless workers, campaign-priority queues and pre-processing of historical CVs provide horizontal scale, while the real-time interview GPU pool remains reserved. Worker capacity will be set to meet the confirmed business window with the agreed headroom.

7.3.1.3 Integration with existing systems and pilot testing

Pilot testing precedes each production release: a controlled group exercises the full path with real data in the staging environment, defects are triaged against the requirements baseline, and the acceptance criteria are verified before the release is proposed for sign-off.

- **Testing Arabic/English CV extraction** — At least 300 per language; electronic/scanned stratified — CER/WER, field P/R/F1, critical exact match, 95% CI — Agreed per-language thresholds and usable confidence/manual-review path
- **Testing Sustained CV throughput** — Real mixed documents; 60 minutes; 1/2/4/8 workers — Pages/min, P95, errors/retries, queue and utilisation — At least business peak with agreed headroom; node-failure capacity verified

7.3.1.4 Deployment and Go-Live

A detailed deployment plan transitions each wave into production. A go-live readiness assessment precedes each release and the system is monitored closely through the transition. Initial post-go-live support is provided on site to address issues and guide stakeholders. All findings arising from OIA's security assessments of the delivered solution are closed before go-live. The assessment windows are scheduled in the project plan and tracked in the dependency register at §7.3; where an assessment is conducted later than the planned window, the affected activity and the release date move by the period of the delay under change control.

7.3.1.5 Post Implementation Support and Training

Ongoing support is provided to internal and external users, covering configuration adjustment and enhancement for optimal performance. A maintenance and monitoring schedule keeps the system updated, secure and fully functional. Support operates 24×7 to prevent integration issues. Training and handover are described at section 8; the support arrangement is described at section 10.

- **Model maintenance** — Continuous monitoring, quarterly review — Drift detection, accuracy and precision tracking, recalibration where indicated, and enhancement of model coverage with additional document types
- **Arabic improvement programme** — Through the support period — Joint programme improving Arabic recognition and parsing accuracy using data OIA provides, with defined review points

7.3.2 Smart Candidate Matching

What it delivers. Two closed loops: an evidence-backed candidate ranking produced when a role enters screening; and a recruiter shortlist or rejection decision with a recorded reason, returned to Jadarah.

Dimensional scoring. Candidates are evaluated against the role across hard skills, soft skills, past role relevance, industry background, job level, years of experience, technical competency, education level, certifications, language proficiency, location and nationality. Every dimension carries a configurable weight and a configurable maximum, set per role or role family. The engine reads the attached CV directly to assess job history against the role, not only the structured profile.

Explainable output. Each candidate receives a match score with the score for every dimension and the percentage achieved. Alongside it the system presents the specific passages in the CV that support the assessment, highlighted in place. Gaps and mismatches are identified explicitly rather than expressed only as a lower score. Scoring is dimensional and traceable by design — the explainability requirement rules out an opaque similarity score, and evidence highlighting is only possible because extraction retains the location of each fact.

Recruiter workbench. A dashboard presents top matches with visual indicators for qualified or disqualified status, the match score, matched attributes expressed as a count against requirements, and experience. From the list the recruiter can open full candidate details, shortlist a candidate — which requires a written reason — or disqualify. The sourcing and shortlisting page carries fixed filters across name, skills, technical competencies, minimum and maximum years of experience, location, job type, industry, career level, minimum degree level, university, minimum GPA, nationality, job-seeker type, professional certificate, current employer and current title. Applicant details export to Excel across the full field set specified in the RFQ.

Matching combines OIA-governed skills/titles taxonomy, deterministic eligibility rules, bilingual lexical search, multilingual embeddings and reranking. Scores are calculated by versioned dimensions and weights and can be recalculated independently. Missing evidence is reported as “not evidenced”, not converted into a negative capability inference. Recruiters see the source CV/JD evidence, gaps and any policy rule that influenced the result before confirming a shortlist or rejection.

If nationality or another policy attribute is required for an approved Omanisation rule, it is implemented as an explicit, configurable and auditable OIA rule. It is not learned as an implicit model feature and does not influence unrelated capability dimensions.

Return to Jadarah. Final results are shared through the integration service.

7.3.2.1 Requirement gathering and design phase

Matching criteria — hard skills, soft skills, experience in years and industry relevance, education level, certifications, language proficiency and location — are defined with OIA in this activity, together with the scoring model: which dimensions are weighted, how, and where a rule applies rather than a score. Deliverables and timing are at section 6.11; the activity closes at MS1 with a signed requirements baseline.

7.3.2.2 Application development and testing

- **Model 2** — Relevance engine, evidence location, ranking and dashboard, filters and export, shortlist workflow
- **Testing Matching and evidence** — At least 300 applications across 3+ job families — Recall@K, NDCG, evidence precision, stability — Recruiter-approved quality; every conclusion reconstructable

7.3.2.3 Integration with existing system and pilot testing

- **Interface 3 - Job data read** — Jadarah → platform — When a role enters screening, and on job update — Job identifier, description, level, requirements — Cached within a defined freshness window; stale reads are flagged rather than used silently
- **Interface 5 - Result return** — Platform → Jadarah — On completion of matching, assessment or interview — Scores, evidence references, report location, status — Queued and retried; every result carries a correlation identifier for reconciliation

7.3.2.4 Deployment and Go-Live

- **Weeks 3-10** — CV parsing plus matching/recommendation — Bilingual extraction results, source traceability, confirmation workflow and initial integration

7.3.2.5 Post Implementation Support and Training

- **Fairness revalidation** — Periodic, per the agreed cycle — Symmetric-pair testing across candidate groups, reported as a named deliverable

7.3.3 Assessment Exams

What it delivers. A closed loop from assessment invitation through to a graded, comparable result returned to the candidate record and to Jadarah.

Test types and content. The platform supports IQ and cognitive tests covering logical reasoning and pattern recognition, multiple-choice questions across general and technical knowledge, and custom assessments built for a role or industry. Item types include multiple choice, case study, essay and practical exercise. Content is provided both as authored banks — cognitive ability, behavioural competency, and discipline-specific banks for engineering, information technology, business and humanities, in Arabic and English — and as authoring tools with which OIA creates and maintains its own content. Question bank management supports creation, editing and deletion, with categorisation by topic and difficulty. Assessments are saved as reusable templates.

Delivery and proctoring. Assessments are delivered online under the full proctoring stack: browser lockdown with fullscreen enforcement and tab-switch detection, webcam monitoring with face-presence and multiple-face detection, AI-based malpractice detection across response timing and answer-pattern anomalies, and human-reviewed flags. Every detection is surfaced to a person for adjudication; none produces an automatic disqualification. Multiple assessment schedules accommodate large candidate volumes, and candidates receive clear instructions and support before and during the assessment.

Grading. Objective items are graded automatically with immediate result calculation, and sections are weighted by relevance to the role. Case study and essay items receive an AI-proposed score with its reasoning and the passages that support it; a person confirms or adjusts before the result is final.

Reporting. An analytics dashboard presents score, accuracy, time taken and benchmark comparison. Individual candidate reports set out performance with strengths and areas for improvement; per-position summary reports aggregate across candidates. Results integrate with the candidate profile and with interview outcomes, and the score and summary are returned to Jadarah.

Authored question-bank volume. The content below is included in the base scope; volumes beyond it are priced as an option.

- **Cognitive ability** — 200 — Logical reasoning, pattern recognition, numerical and verbal reasoning, across three difficulty levels
- **Behavioural competency** — 150 — Situational judgement and behavioural scenarios mapped to the evaluation dimensions
- **Engineering** — 150 — Multiple choice, case study, essay and practical items across three difficulty levels
- **Information technology** — 150 — As above
- **Business** — 150 — As above
- **Humanities** — 150 — As above
- **Total**: The committed question bank comprises 950 source items, delivered in Arabic and English as 1,900 language versions. Requirements sign-off freezes the item distribution, discipline mix, difficulty profile and bilingual counting convention; the base quantity changes only through formal change control.

Ownership and reuse. Items authored for OIA are delivered as OIA's content. OIA may edit, extend, retire and reuse them without restriction and without reference to the supplier. The authoring tools delivered with this model allow OIA to maintain and grow the bank after handover.

Arabic quality. Every Arabic item is reviewed by a native speaker for language quality and, separately, for equivalence with its English counterpart — an item must test the same thing in both languages rather than being a translation. Review records are delivered with the bank.

Item security. Assessments are assembled at random from the bank against a blueprint rather than served as a fixed paper; item exposure is tracked so that over-exposed items are rotated out; and the bank supports versioning so that retired items remain auditable against past results.

Question sources are managed in four distinct categories: OIA competency and job material; original subject-matter-expert items; properly licensed content; and validated, normed psychometric instruments from licensed providers. LLMs may support drafting and controlled variant generation but do not become the source of record.

Every item records provenance, licence, author, reviewer, competency, difficulty, answer/rubric, Arabic equivalence, version, exposure and retirement status. Arabic versions use translation, back-translation and native subject-matter review to confirm measurement equivalence rather than literal similarity.

Custom job questions will not be described as normed IQ, personality or cognitive instruments. Where normed assessment is required, an industrial/organisational psychologist and the licensed instrument provider must confirm reliability, validity, differential item functioning, norm applicability and accommodations.

7.3.3.1 Requirement gathering and design phase

G10 — Assessment — Adequate pilot sample per item class — Reliability, difficulty/discrimination, DIF, equivalence — Qualified psychometric sign-off for the represented use

7.3.3.2 Application Development and Testing

Model 3 — Exam design and development across technical, aptitude and behavioural competencies with industry-relevant and real-world items, customised by discipline, spanning multiple choice, case study, essay and practical formats; exam administration with secure proctoring, multiple schedules and candidate support; evaluation and scoring with automated grading, weighted sections and detailed reports; results and reporting with individual scores, ranking lists, strengths and improvement areas, and reports for hiring managers, held under confidentiality and secure storage
The assessment service includes question authoring and review, job/competency blueprints, bilingual equivalence review, randomised assembly, server-side timing, event-based answer recovery, deterministic scoring for objective items and rubric/evidence-supported review for subjective items. LLMs may draft content or suggest a score, but an approved question version and rubric remain authoritative.

Untrusted live-coding submissions execute in a separate node pool using kernel-sandboxed or hardware-isolated execution, non-root and read-only filesystems, deny-all networking, strict CPU/memory/process/time/output limits, fixed signed language images and disposable execution environments.

Candidate code is treated as hostile. Execution is separated from application and data nodes and uses a dedicated sandbox runtime: a gVisor userspace-kernel sandbox for compatible high-density workloads or a Kata Containers lightweight-VM runtime where hardware-backed isolation is preferred. Both paths use deny-all networking, no access to Jadarah, the orchestration control plane or internal DNS, non-root users, read-only base images and resource/time/output limits. Images are fixed, signed and available offline. Each environment is destroyed after use and only the permitted code hash, resource evidence and score audit are retained. Final runtime selection follows compatibility, performance and adversarial containment testing under Gate G13.

7.3.3.3 Integration with existing system and pilot testing

- **G13** — Code sandbox — Escape, fork bomb, network and filesystem tests — Containment, limits, cleanup and audit — No host/internal-network access; reliable termination and disposal

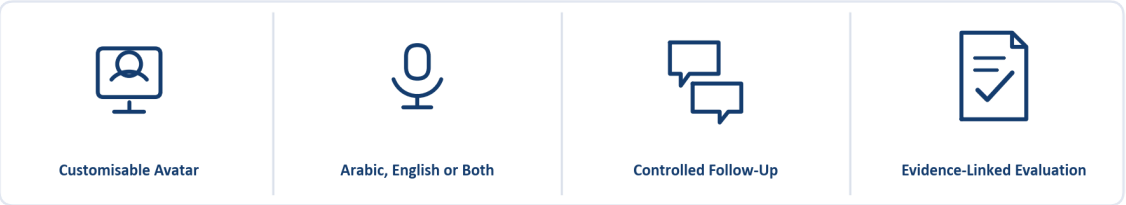
All proctoring events remain evidence flags for human review. Candidate consent, purpose limitation, retention, deletion, alternative verification and reasonable accommodations are part of the go-live gate.

7.3.3.4 Deployment and Go-Live

- **Weeks 5-17** — Assessment — Agreed bank scope, authoring, delivery, scoring, proctoring and psychometric review

7.3.3.5 Post Implementation Support and Training

- **Assessment authors** — Question bank management, categorisation, exam template construction, blueprint-based assembly, item review workflow — One half-day session



7.3.4 AI Interviewer

What it delivers. Three closed loops: identity confirmation at session start; a two-way interview with dynamic follow-up questioning; and a per-dimension evaluation with evidence and an outcome recommendation. The module operates end to end — from the greeting through to the recommendation placed before the recruiter — with the hiring decision resting with a person.

Session creation and access. HR configures the interview in Jadarah: job type, job description, interview duration, preset static questions and the evaluation dimensions with their weights. The platform returns a single-use, time-bounded unique URL, which Jadarah distributes to the candidate. The page the candidate reaches may carry multiple assessments in sequence.

Identity confirmation: At session start the candidate presents an identity document to the camera and a live capture is matched against it, with a liveness check. The reference image comes from the document itself, so the design carries no dependency on photographic data existing elsewhere. Where the match is uncertain or fails, the interview proceeds normally and the session is flagged for HR adjudication using the recording and the captured document. A recognition error costs a review, never a candidate. A device pre-check — camera, microphone, lighting, connectivity — runs before the session begins.

Conversation: The interview opens with a greeting supplied by OIA and customised by job level. The avatar asks the preset questions and, based on what the candidate actually says, asks follow-up questions. Follow-up generation is controlled: for each question HR's configuration defines the dimensions along which the interview may probe, and the system generates within that scope behind topic guardrails, with the full exchange logged. This satisfies the requirement for dynamic two-way dialogue while preventing drift and — importantly for the fairness requirements — making it impossible for the line of questioning to differ

systematically between candidate groups. Multiple-choice, case study, essay and practical items may be interleaved with the conversation, and live coding with sandboxed execution is available for technical roles. The voice pipeline is fully streaming and handles interruption: if the candidate begins speaking while the avatar is talking, playback stops and the system listens.

Language: Interviews run fully in English, fully in Arabic, or with the two mixed within a single session. Where recognition confidence on a passage is low the interviewer asks the candidate to restate — natural in a real interview — and where confidence across a session falls below threshold the transcript is flagged and the assessment routed to human review rather than scored down. A text-based interview channel using the same questions, follow-up logic and scoring dimensions is available as an equivalent alternative, and live captioning accompanies the spoken interview.

The architecture supports illustrated 2D, rigged 3D and photorealistic avatar classes. The default implementation uses deterministic phoneme-to-viseme control for an approved 2D/3D rig because it provides predictable latency, operation and recovery. A locally deployable generative lip-synchronisation path may be added after its dependencies, model weights, visual assets and performance have been approved.

The acceptance test covers Arabic, English and code-switch clips, including long vowels, emphatic/pharyngeal sounds, different speeds and pauses. It reports effective frame rate, audio/video offset, first-frame time, GPU use, native-speaker acceptability and time to fall back to static-avatar/audio or text-only operation.

Avatar rendering is outside the scoring path and cannot delay the first audible response. The fallback order is full avatar, an approved static image with synchronised audio and captions, then a text-only interview using the same question and scoring state.

Avatar: The interviewer is presented as a customisable avatar built to the design OIA supplies, with real-time lip-sync in both languages, a defined repertoire of facial expressions and gestures driven by dialogue state, and contextually appropriate responsiveness — expression, tone and pacing adapting to the interview stage and to the candidate's response state, for instance a supportive pause where a candidate hesitates. Rendering runs entirely on local infrastructure. Under load the session degrades in a defined sequence rather than failing.

Evaluation: After the session the system produces a per-dimension evaluation against the criteria HR configured, with each score supported by verbatim quotation from the transcript. Behavioural attributes are reported per attribute — tone of voice as speech rate, pause pattern, volume variation and fluency with timestamped audio references; eye contact as camera-directed gaze proportion over the session; facial expression as observed categories and their distribution; body language as posture stability and gesture activity — each indicator resolving to the exact segment that produced it. Composure and assertiveness are reported as behavioural dimensions evidenced by response content and interaction pattern.

The multimodal pipeline uses dedicated audio, video and language feature extractors followed by temporal aggregation and calibrated cross-modal fusion. It can assess RFQ-requested observable indicators including speech rate, pauses, volume, fluency, gaze, facial expression, posture, gesture, composure/calmness, assertiveness, aggression indicators and response to challenge. Personality-related dimensions, where requested, combine validated assessment instruments, structured linguistic evidence and longitudinal behavioural patterns; they are not inferred from a single facial expression or acoustic event. Each result exposes the contributing evidence, confidence and model/rule version, is tested for subgroup performance and remains subject to authorised human review. No single biometric feature can automatically reject or shortlist a candidate.

Scale and administration. Multiple interviews run concurrently. Scheduling and reminders are automated. Results export to PDF and Excel and return to Jadarah through the integration service.

The interview service provides device checks, OIA-hosted WebRTC connectivity, VAD/endpointing, streaming ASR, controlled follow-up, local LLM inference, Arabic/English TTS, captions, avatar synchronisation, interruption handling, multimodal behavioural analysis and active-session continuity. The session orchestrator, not the LLM, owns question order, allowed dimensions, maximum follow-ups, policy controls and completion state.

Interview scoring uses job-related behavioural anchors, transcript evidence and time-series multimodal features. Where required by the RFQ and approved by OIA, dedicated models analyse speech rate, pause structure, volume consistency, fluency, gaze direction and proportion, facial expression patterns, posture, movement and gesture. These observations support calibrated dimensions such as composure/calmness, assertiveness, aggression indicators and response to challenge. Job-relevant personality dimensions are assessed through validated questionnaires, structured responses and longitudinal behavioural evidence rather than a single frame or signal. Results remain explainable, fairness-tested and subject to authorised human confirmation. Identity and proctoring exceptions create review flags, not automatic disqualification.

7.3.4.1 Requirement gathering and design phase

HR configuration defines fixed questions, allowed follow-up dimensions, prohibited topics and maximum follow-ups. The orchestrator passes structured state and requires structured output such as action, question, target dimension, supporting evidence and safety flags. Model, prompt, sampling parameters and job configuration are versioned.

7.3.4.2 Application Development and Testing

Core interviewing system supporting simultaneous multi-candidate interaction; interfaces for recruiters and candidates; automated scheduling and real-time evaluation; HR configuration of job types, descriptions, durations and static questions with AI generating additional questions from the job description; candidate validation filters including skill assessment and behavioural analysis, customisable to role and industry; scalability and performance optimisation for large-scale operation; human-like interaction with natural language understanding, context-aware response, conversational depth and adaptive follow-up

Implementation: NVIDIA Nemotron 3.5 ASR Streaming 0.6B is the primary reference for incremental captions and low-latency finalisation. An equivalent cache-aware native-streaming Arabic/English recogniser may be selected if it meets the same tests. Whisper large-v3/turbo provides an independent multilingual comparison and fallback, with an Arabic-specialised final-transcript path for difficult Gulf/Omani and code-switching conditions.

Test data: A speaker-disjoint 10-20 hour blind set covering Omani regions, gender and age mix, MSA/Gulf variation, English-Arabic code-switching, job terms, names, numbers, dates, common devices and realistic noise. Training, development and test speakers must not overlap.

Published Gulf Arabic conversational corpora and Omani speech resources will support vocabulary analysis and comparative benchmarking, subject to a rights review. Because those sources do not reproduce OIA browser interviews, devices, noise or job terminology, the project will also create a dedicated OIA acceptance set with documented consent, usage rights and native-speaker transcription.

Metrics: Raw and normalised WER, critical-entity error rate, partial stability, finalisation delay, real-time factor and end-to-end turn latency, split by Arabic-only, English-only and code-switch conditions.

Fallback: Low-confidence speech triggers repetition or the equivalent text channel; repeated ASR uncertainty does not reduce a candidate score. The selected configuration may combine native streaming, authorised in-domain adaptation, an independent multilingual recogniser and an Arabic-specialised stable-transcript path without changing the external interface, orchestration or scoring controls.

The speech chain comprises Arabic text normalisation, optional Tashkeel/diacritisation, an OIA pronunciation lexicon, pronunciation overrides, locally deployable bilingual neural TTS and phoneme/viseme timing. SILMA-TTS is the named bilingual reference, with a CATT-class locally deployable diacritisation path evaluated against direct synthesis. OIA will approve the production voice against Omani/Gulf intelligibility, pronunciation, latency and usage rights. The pronunciation lexicon covers names, OIA terms, locations, job titles and English abbreviations.

Production approval covers both voice quality and usage rights. The project will document permitted commercial, on-premise, language, territory, duration, derivative and retraining uses. If no suitable pre-cleared voice is available, UIS Cloud will commission a consented Omani/Gulf voice pack.

Validation uses 150-200 representative sentences and blind review by 10-15 target-region native speakers. It reports intelligibility, MOS, pronunciation errors, code-switch errors, first-audio latency and real-time factor, comparing direct TTS with the diacritisation/lexicon path.

Pronunciation is managed through diacritisation, the lexicon and explicit overrides. If the selected voice does not meet the pronunciation or rights requirements, the fallback order is an alternative approved local voice, a commissioned and consented voice pack, and an accessible text/caption channel until the approved voice is available.

The primary service metric is measured from client-observed end of candidate speech to the first audible TTS sample on that same client. One distributed trace links VAD endpoint, stable ASR final, orchestration, LLM first token, TTS first chunk, avatar first frame and client playback.

Tests cover at least 200 representative turns at 1, 5 and 10 concurrent sessions and report client-observed and server-only P50/P95/P99 values for English, Arabic and code-switching, plus timeout, retry, ASR revision, fallback and interruption rates.

Targets: English P95 first audible response at or below 2.5 seconds; Arabic at or below 3.0 seconds; P95 interruption stop at or below 300 ms, all at ten concurrent sessions. The avatar must not block audio response.

The reference model set uses Qwen3-14B for general bilingual instruction following and structured output, with Jais-2-8B-Chat as the Arabic-focused comparison and approved alternative. Jais-2-70B-Chat may be used offline as a quality comparator or teacher model; it will enter the live N+1 serving configuration only if a production-equivalent test confirms the required concurrency and latency. The live model will be selected on Arabic interview quality, structured-output compliance, prohibited-topic rate, time to first token, GPU utilisation and usage rights.

All scoring is traceable to a rubric or behaviourally anchored rating scale and the supporting transcript, audio interval, video interval, question and feature version. The language model proposes a structured explanation; deterministic rules and approved scoring configurations calculate the assessment dimensions; authorised users confirm or override the result with a recorded reason.

7.3.4.3 Integration with existing system and pilot testing

Each wave integrates with Jadarah and with OIA's HR tooling, and is deployed to a secure environment with data protection measures compliant with applicable regulation. For Model 4 specifically, the interview process is established from the Jadarah platform: HR creates the job detail there, the AI platform returns the URL to Jadarah, and the unique URL directs the jobseeker to a page which may include multiple assessments.

- **Interface 4** — Interview creation and unique URL return — Bidirectional — HR creates an interview in Jadarah — Inbound: job, duration, preset questions, evaluation dimensions. Outbound: single-use, time-bounded session URL — URL generation is synchronous; on failure the creation is rejected with an explicit error rather than left pending
- **G6** — End-to-end turn — At least 200 turns at 1/5/10 sessions — Client/server P50/P95/P99, fallback, barge-in — Acceptance target: 2.5 s EN / 3.0 s AR P95; 300 ms barge-in at ten sessions
- **G7** — Avatar — Arabic/English/mixed clips at ten sessions — FPS, A/V offset, first frame, GPU, blind review — At least 25 FPS where applicable; accepted quality and immediate fallback
- **G8** — Controlled dialogue and behavioural analysis — At least 200 dialogue simulations plus representative audio/video sessions and red-team set — Follow-up relevance, prohibited-topic rate, structured-output compliance, dimension accuracy, calibration and evidence traceability — Zero prohibited-topic follow-up; agreed dialogue and behavioural thresholds; every result evidence-linked and human-reviewable
- **G12** — Proctoring — Normal, anomalous and accommodation scenarios — Detection P/R, false flags and review load — No automatic rejection; complete evidence and accommodation path.
  Face matching and presentation-attack detection, if approved, return “pass” or “manual review”; they do not automatically cancel eligibility. Testing covers genuine/impostor attempts, printed images, screen/video replay and representative lighting, devices and demographic groups.
  The solution provides two levels of proctoring. Browser controls detect tab visibility changes, full-screen exit, camera and microphone interruption, face absence/multiple-face events, copy/paste and anomalous interaction patterns. For assessments classified by OIA as high-stakes, UIS Cloud supports a managed secure-assessment client or OIA-managed kiosk policy with application allow-listing, restricted navigation, process/device controls and restricted launch/exit. The selected level is recorded in each assessment blueprint and tested under Gate G12.
  All proctoring events remain evidence flags for human review. Candidate consent, purpose limitation, retention, deletion, alternative verification and reasonable accommodations are part of the go-live gate.
- **G14** — HA and active-session continuity — Application, media, GPU process/node, database and network failures — Automatic rerouting, session continuity, state reconstruction, data loss and RPO/RTO — Interview survives without candidate restart; no confirmed turn lost; agreed recovery time demonstrated
- **G18** — Jadarah integration — Contract, sandbox, retry, outage and reconciliation scenarios — Versioned I1-I5 contracts, end-to-end traces and reconciliation report — Agreed records exchange correctly; retries are idempotent and outages recover without silent loss

7.3.4.4 Deployment and Go-Live

- **Weeks 5-20** — Speech, interview and avatar — Gulf/Omani ASR, TTS, dialogue, latency, sandbox, avatar and text fallback evidence

7.3.5 Application Scoring

Model 5 — Candidate-facing scoring, derivation display, recommendation generation, administrator configuration, activity logging
What it delivers: Two closed loops facing the candidate: a match score with its derivation and improvement recommendations when the candidate applies to or expresses interest in a role; and open roles ranked against the candidate's profile.

Scoring and explanation: The score is produced by the same matching component that serves the recruiter view, assessed across relevant experience, relevant skills, relevant industry and relevant past positions. The candidate is shown how the score was derived — which requirements were matched, which were partially matched, and which were not evidenced in the profile.

Recommendations: Improvement recommendations are structured and actionable: what is missing relative to the role, and what evidence would strengthen the application. Generated text is constrained to verifiable facts drawn from the profile and the job description. The system makes no evaluative statement about the person — a constraint that protects the candidate and equally protects OIA, since every statement made to an applicant is one that can be substantiated.

Reverse recommendation: Against the candidate's profile and CV, currently advertised roles are ranked from high to low match and presented with the reason for each.

- **CV matching and job recommendation** — Comply — Shared bilingual taxonomy, hybrid retrieval, versioned weights, evidence-backed scoring and recruiter/candidate explanations under Gates G9 and G17
- **G17** — Fairness — Counterfactual, subgroup and operational data — Score/error/selection gaps with confidence intervals — No unexplained material disparity; governance approval
  The contractual requirement is WCAG 2.0 AA, with WCAG 2.2 AA used as the engineering target. Testing covers the complete candidate, recruiter and administrator journeys, including keyboard-only use, focus, screen readers, zoom/reflow, contrast, Arabic RTL and mixed bidirectional text, error handling, exported reports, captions and equivalent text participation.
  Fairness assurance combines counterfactual pairs, language-specific error analysis, subgroup and intersectional results, selection-rate diagnostics, differential item functioning, human override and appeal patterns. Any material unexplained disparity triggers investigation and controlled revalidation. Model, rule, weight or question-bank changes cannot bypass the release gate.

7.4 Technology Stack

The following products form the reference stack used for implementation, sizing and acceptance planning. A component may be replaced only if the substitute meets the same functional, performance, security, offline-deployment and licensing requirements. Exact versions, hashes, dependencies and licence obligations will be recorded in the confidential SBOM, Model BOM and asset registers.

Capability	Reference stack	Alternative / purpose	Selection and acceptance criteria
Document intelligence	PaddleOCR-VL-1.6 plus PP-OCRv5 Arabic	Equivalent Arabic-capable OCR/layout model for comparative validation	G1 Arabic/English accuracy and confidence; G2 complete-chain throughput; local/offline rights
Matching and retrieval	Bilingual lexical/vector retrieval, multilingual embeddings and reranking	Independently replaceable embedding/reranking path with registered taxonomy versions	G9 quality, evidence precision, stability and explainable policy rules
Controlled language generation	Qwen3-14B	Jais-2-8B-Chat for Arabic-focused comparison; Jais-2-70B-Chat as an offline quality/teacher reference	G6 latency and N+1 capacity; G8 structured-output, Arabic quality and policy compliance
Streaming speech recognition	Nemotron 3.5 ASR Streaming 0.6B	Whisper large-v3/turbo plus an Arabic-specialised stable-transcript path	G3/G4 Gulf/Omani and code-switch evidence, partial stability, RTF and latency
Arabic speech synthesis	SILMA-TTS, diacritisation and OIA pronunciation lexicon	Commissioned Omani/Gulf voice pack and alternative locally deployable TTS	G5 intelligibility, pronunciation, quality, first-audio latency and voice rights
Avatar rendering	Deterministic phoneme-to-viseme 2D/3D renderer	Locally deployable generative lip-synchronisation enhancement	G7 frame rate, synchronisation, visual quality, asset rights and immediate fallback
Behavioural analytics	Dedicated audio/video/language feature models with temporal cross-modal fusion	Separately calibrated small models per observable dimension	G8 interpretability and G17 subgroup, fairness and human-review evidence
Secure code execution	gVisor-class userspace-kernel sandbox or Kata Containers-class lightweight VM	Hardware-isolated disposable worker pool for incompatible/high-risk workloads	G13 containment, compatibility, limits, cleanup and adversarial testing
Platform and audit	Kubernetes-compatible orchestration, vLLM-class local serving, OpenTelemetry-compatible tracing, PostgreSQL/pgvector-class data services	OIA-approved equivalent supported components	G14-G16 resilience, security, air-gap, supportability and supply-chain records

7.5 Technical Risk and Mitigation Register

ID	Risk area	Risk	Mitigation and closure	Owner
A01	Arabic CV extraction	Accuracy varies materially by scan quality and layout	G1 blind bilingual report; confidence and confirmation workflow	Document AI Lead
A02	Gulf/Omani ASR	Public Arabic results may not represent Omani interviews	G3/G4 dedicated blind set; model comparison; text fallback	Speech Lead
A03	Arabic voice quality and rights	A technically available voice may lack local acceptability or production rights	G5 native review; pronunciation lexicon; signed voice register	Speech Lead + Native QA

ID	Risk area	Risk	Mitigation and closure	Owner
A04	Ten-session real-time performance	Combined ASR/LLM/TTS/avatar may exceed latency or memory budget	G6/G7 production-equivalent load test; isolated real-time pool	Performance Lead
A05	Question-bank validity and distribution	Discipline mix, rights or psychometric claims may require refinement	Committed 950/1,900 base; frozen distribution; provenance register; qualified psychometric review	Assessment Lead
A06	Jadarah readiness	Late API/sandbox access affects integration critical path	Week-2 contract gate, mock service and dependency escalation	Integration Lead + OIA
A07	Licence and asset rights	A code licence does not cover every model/data/voice/avatar asset	G15 SBOM and asset registers; alternative component path	Legal/Security Lead
A08	Active-session failover	A node fault may cause a brief media pause while volatile context is reconstructed	N+1 capacity, redundant media, pre-warmed inference, journalled state, automatic reconnection and G14 fault injection	Platform Lead
A09	High-assurance proctoring	Endpoint policy and secure-client deployment must match assessment risk	Risk-tiered controls, managed secure-assessment client/OIA kiosk policy and G12 adversarial test	Assessment/Security Lead
A10	Untrusted code	Ordinary application containers are not a sufficient security boundary	gVisor/Kata-class sandbox baseline, hardware-isolated alternative and G13 adversarial tests	Security Lead
A11	Candidate rights and fairness	Multimodal behavioural processing can create privacy and bias exposure	DPIA, evidence-linked dimensions, calibration, subgroup testing, human confirmation, G17 and accommodation path	AI Governance + OIA HR
A12	Hardware procurement	Purchasing before model/load evidence can under- or over-size the estate	Exact-SKU benchmark, N+1 validation and signed sizing report	Solution Architect + Infrastructure Lead

7.6 Verification and Acceptance Framework

Gate	Verification object	Representative data/load	Mandatory evidence	Proposed exit condition
G1	Arabic/English CV extraction	At least 300 per language; electronic/scanned stratified	CER/WER, field P/R/F1, critical exact match, 95% CI	Agreed per-language thresholds and usable confidence/manual-review path
G2	Sustained CV throughput	Real mixed documents; 60 minutes; 1/2/4/8 workers	Pages/min, P95, errors/retries, queue and utilisation	At least business peak with agreed headroom; node-failure capacity verified
G3	Gulf/Omani ASR	10-20 hours, speaker-disjoint	Raw/normalised WER, critical entities, RTF, final delay	Business-acceptable critical entities and agreed improvement over baseline

Gate	Verification object	Representative data/load	Mandatory evidence	Proposed exit condition
G4	Code-switch ASR	At least 2 hours of realistic Arabic-English speech	Per-language WER, switch-point error, partial stability	Degradation remains within the jointly agreed tolerance
G5	Arabic TTS	150-200 sentences, native blind review	MOS, intelligibility, pronunciation errors, RTF	Native-speaker and OIA terminology gate passed
G6	End-to-end turn	At least 200 turns at 1/5/10 sessions	Client/server P50/P95/P99, fallback, barge-in	Acceptance target: 2.5 s EN / 3.0 s AR P95; 300 ms barge-in at ten sessions
G7	Avatar	Arabic/English/mixed clips at ten sessions	FPS, A/V offset, first frame, GPU, blind review	At least 25 FPS where applicable; accepted quality and immediate fallback
G8	Controlled dialogue and behavioural analysis	At least 200 dialogue simulations plus representative audio/video sessions and red-team set	Follow-up relevance, prohibited-topic rate, structured-output compliance, dimension accuracy, calibration and evidence traceability	Zero prohibited-topic follow-up; agreed dialogue and behavioural thresholds; every result evidence-linked and human-reviewable
G9	Matching and evidence	At least 300 applications across 3+ job families	Recall@K, NDCG, evidence precision, stability	Recruiter-approved quality; every conclusion reconstructable
G10	Assessment	Adequate pilot sample per item class	Reliability, difficulty/discrimination, DIF, equivalence	Qualified psychometric sign-off for the represented use
G11	Identity/PAD	Genuine/impostor plus presentation attacks	FMR/FNMR, APCER/BPCER and subgroup results	Failures route to human review; security/privacy approval
G12	Proctoring	Normal, anomalous and accommodation scenarios	Detection P/R, false flags and review load	No automatic rejection; complete evidence and accommodation path
G13	Code sandbox	Escape, fork bomb, network and filesystem tests	Containment, limits, cleanup and audit	No host/internal-network access; reliable termination and disposal
G14	HA and active-session continuity	Application, media, GPU process/node, database and network failures	Automatic rerouting, session continuity, state reconstruction, data loss and RPO/RTO	Interview survives without candidate restart; no confirmed turn lost; agreed recovery time demonstrated
G15	Air-gap and licences	24-hour full journey with external egress disabled	Egress, activation, telemetry and complete BOM/registers	Zero unapproved external calls; no unresolved usage-right gaps
G16	Security	ASVS/API, SAST/DAST/SCA and penetration test	Findings and remediation evidence by severity	All go-live findings closed or formally accepted by OIA
G17	Fairness	Counterfactual, subgroup and operational data	Score/error/selection gaps with confidence intervals	No unexplained material disparity; governance approval
G18	Jadarah integration	Contract, sandbox, retry, outage and reconciliation scenarios	Versioned I1-I5 contracts, end-to-end traces and reconciliation report	Agreed records exchange correctly; retries are idempotent and outages recover without silent loss
G19	Accessibility	Complete candidate/recruiter/admin journeys	Automated, manual and assistive-technology findings	WCAG 2.0 AA contractual pass; 2.2 AA engineering target

7.7 Information Security Requirements

The deployment is designed against the four annexed information security requirement documents — API Security, Web Application Development, Hosting Service Provider, and Supplier Information Security. The clause-level response is at Annex E. In summary: network segmentation between interface, application and data tiers; a web application firewall protecting the candidate-facing boundary; intrusion detection on the perimeter; encryption in transit and at rest; role-based access with multi-factor authentication for privileged accounts; secure development practices with security testing in the pipeline; and independent penetration testing before go-live, with all findings in the delivered solution remediated before release.

- Kubernetes-based on-premise deployment with separate real-time and batch resource pools.
- HA relational and vector data services, encrypted object storage, reliable session/event services and append-only audit evidence.
- OIA-controlled keys, secrets, certificates, encryption at rest and in transit, network segmentation and deny-by-default egress.
- Offline OCI/package/model repositories with signed artefacts, hashes, provenance and controlled promotion.
- End-to-end distributed traces, metrics and logs integrated with the OIA SIEM; recruitment decision evidence is retained separately from routine operational logs.
- Model registry and release gates covering quality, fairness, security, performance, licence and rollback evidence.
- Backup, restore and disaster-recovery tests covering database, objects, configuration, models, keys and audit evidence.

The RFQ requires a plan covering the entire lifecycle of data and model training, with anti-tampering, privacy protection, encryption, auditing and data-sovereignty protection at every stage. The table below states, for each stage, where the data sits and what protects it.

- **Extraction** — CV upload, candidate profile read from Jadarah, session audio and video capture — In Oman — TLS in transit; token-validated access; sensitive field identification on ingestion
- **Processing** — Document parsing, transcription, evaluation, scoring — In Oman, local GPU — Processing confined to the application network segment; no egress route configured
- **Transmission** — Results returned to Jadarah; internal service-to-service calls — Internal network only — Mutual TLS; per-service tokens; API controls per the OIA API Security Requirements
- **Training and tuning** — Calibration on OIA-provided historical data — In Oman, on the same infrastructure — Training data masked of direct identifiers; no data leaves the environment for training at any point
- **Inference** — All model execution — In Oman, local GPU — No external inference endpoint exists in the configuration
- **Storage** — Profiles, recordings, transcripts, evaluations, audit records — In Oman — Encryption at rest; role-based access; sensitive data masking in non-production environments
- **Retention and deletion** — Configurable retention policy per data class, with verified deletion — In Oman — Policy engine with audit trail of deletion events; default periods set at requirements sign-off

Anti-tampering and audit. Every decision record — inputs, applied rules, AI recommendation and human confirmation — is written to an append-only audit log with integrity verification. Records are retained for the period OIA specifies and are exportable for regulatory review, in accordance with the MTCIT General Policy for the Safe and Ethical Use of Artificial Intelligence Systems.

Sovereign deployment fails in practice when a component quietly requires the outside world. The following are stated as design constraints and verified at acceptance:

- No component calls a cloud AI service for inference, transcription, synthesis or rendering
- No component requires online licence activation or periodic licence revalidation; all licences are activated offline following production deployment
- No component transmits telemetry, usage statistics or diagnostic data outside the environment
- No component requires an external service for identity, time synchronisation or certificate validation beyond what OIA already operates
- The environment operates correctly with the external network boundary closed to outbound traffic, and this is demonstrated as an acceptance test

Model provenance. All AI models are obtained as open-weight artefacts and deployed locally. Model files are transferred once, at installation, through OIA's controlled process; thereafter the environment operates without any model download, update call or telemetry to an external service. Model updates during the support period are delivered as controlled packages through the same process, at OIA's discretion.

Related to the Security, licensing and offline supply chain, The deliverable will include an SPDX or CycloneDX SBOM, Model BOM and Dataset/Question/Voice/Avatar/Font Asset Registers. Each record identifies version or commit, hash, source, licence, obligations, intended use and approval. Signed OCI/model artefacts and offline OS/Python/model repositories support reproducible installation without public downloads.

Every delivered component will have documented rights for OIA's commercial, on-premise and offline use throughout the intended lifecycle. The control objective is complete and compatible usage rights rather than one licence category across all assets. Non-commercial, research-only or otherwise incompatible assets are excluded unless an appropriate commercial licence is procured and approved.

- **G15** — Air-gap and licences — 24-hour full journey with external egress disabled — Egress, activation, telemetry and complete BOM/registers — Zero unapproved external calls; no unresolved usage-right gaps
- **G16** — Security — ASVS/API, SAST/DAST/SCA and penetration test — Findings and remediation evidence by severity — All go-live findings closed or formally accepted by OIA

7.8 Access Management Requirements (3%)

This section responds to Evaluation Criterion 4 and to the Access Management Requirements at RFQ section 6.1.

Two distinct populations use the solution and they are authenticated differently.

- **Internal — recruiters, administrators, compliance reviewers** — Single sign-on against OIA's own identity provider, with multi-factor authentication enforced. The solution holds no separate internal credential store; account lifecycle follows OIA's existing joiners, movers and leavers process
- **Candidates** — Identity is issued and managed by Jadarah, as confirmed in clarification. The platform receives and validates a Jadarah-issued token and creates no separate candidate credential. Identity confirmation at interview start is a second factor within the session, not a primary authentication
- **Service accounts** — Machine identities for service-to-service and integration calls, authenticated by mutual TLS with per-service tokens, scoped to the minimum required operations and rotated on a defined schedule

Access to functionality, data sets and information context is governed by role. Roles are configurable; the set below is proposed and confirmed at requirements sign-off.

- **Super Administrator** — System configuration, scoring weights, skill taxonomy, model parameters, maximum score per parameter, retention policy, role assignment. Five to ten users as stated in the RFQ sizing
- **Recruiter** — Candidates and roles within assigned scope; ranking, evidence, shortlist and rejection decisions; assessment and interview configuration and results
- **Hiring manager** — Read access to candidates and results for their own roles; comment and recommendation without decision authority unless granted
- **Assessment author** — Question bank creation, editing and categorisation; exam template management; no access to candidate records
- **Compliance reviewer** — Audit records, fairness validation reports and decision reconstruction; read-only across candidate data
- **Support operator** — Operational monitoring and diagnostics; access to personal data restricted and logged, with masked views by default
- **Candidate** — Own profile, own applications, own results and recommendations only

Access follows least privilege. Privileged actions — configuration change, role assignment, retention change, bulk export, access to unmasked personal data — are logged with the actor, the time and the change made, and are reportable to OIA's compliance function. Non-production environments operate on masked data and carry no route to production records.

- Responsive candidate, recruiter and administrator portals with English LTR and Arabic RTL behaviour.
- CV upload, document preview, save/resume, field confirmation and evidence navigation.
- OIA SSO through OIDC or SAML, MFA for authorised internal users, role-based access control and segregation of duties.
- Jadarah-issued, single-use and time-limited candidate access tokens.
- Accessible online assessment and AI interview journeys with captions, keyboard use, assistive-technology support and an equivalent text channel.
- Permission-controlled dashboards and Arabic-safe PDF/XLSX exports.

7.9 Business Analysis Requirements (10%)

This section responds to Evaluation Criterion 6 and to the Business Analysis Requirements at RFQ section 7.

Business analysis on this engagement is not a documentation exercise that precedes build. It is the mechanism by which OIA's recruitment process is turned into a structured, executable and auditable model — the same model the system then runs on. Every requirement gathered becomes an entity, a rule, a use case or an acceptance criterion in that model, and every one of them remains traceable from the stakeholder who raised it through to the test that verifies it.

A dedicated Business Analyst is assigned to the engagement for its full duration and is present at OIA as required throughout, in line with RFQ 7.4. The Business Analyst's curriculum vitae is at Annex D.

RFQ 7.3 requires detailed requirements to be gathered from key stakeholders including IT, HR and Admin. The engagement plan below is proposed and confirmed at kick-off.

- **HR and recruitment** — Current recruitment workflow end to end; screening criteria and how they vary by role family; interview practice; assessment practice; decision authority and escalation; reporting needs — Two structured workshops in weeks 1–2, followed by weekly working sessions through requirements
- **IT** — Jadarah architecture and interfaces; identity provider and access model; hosting environment and network boundary; security baseline and assessment process; deployment and change procedures — Workshop in week 1; technical working sessions through weeks 2–4

- **Admin and business support** — Administrative workflow around recruitment; document handling; record-keeping obligations — Workshop in week 2
- **Compliance and data protection** — Data governance obligations; masking requirements; retention policy; MTCIT AI policy conformance expectations; audit and review process — Workshop in week 2; review checkpoint before each release
- **Candidate representative view** — Candidate experience expectations, accessibility needs, communication tone — Session in week 3

RFQ 7.2 and the supporting requirements call for a thorough analysis of existing systems, processes and workflows, identifying opportunities for improvement and ensuring the new solution integrates with current operations. This is delivered as:

- Current-state process model — the recruitment workflow as it operates today, from requisition through to offer, with the actors, systems, handovers and decision points at each stage, and the time and effort each consumes
- System and data landscape — Jadarah and any adjacent systems in scope, their data, their interfaces and their constraints, including where recruitment data currently lives outside systems
- Pain point register — where effort concentrates, where consistency is hardest to maintain, and where decisions currently lack an evidence trail

RFQ 7.8 requires a detailed gap analysis report identifying current challenges and how the proposed solution addresses them. The report is delivered in week 3 and structured so that every gap traces forward to a specific capability and backward to a specific observation:

- **Gap** — The current-state limitation, stated concretely
- **Evidence** — The workshop record, system observation or data point establishing it
- **Business impact** — Effect on effort, consistency, candidate experience or defensibility
- **How the solution addresses it** — The specific use case or capability, referenced by identifier
- **Residual gap** — What remains unaddressed and why — stated rather than omitted
- **Measure** — How OIA will know the gap has closed

A single requirements baseline is produced and signed off at MS1, covering functional requirements, non-functional requirements, user stories and use cases as required. Each requirement carries a unique identifier, a source reference to the stakeholder or RFQ clause it derives from, a priority, and an acceptance criterion.

Use cases are specified as closed loops — trigger, actor, execution steps, outcome, the point at which a person confirms, and the criteria by which the loop is judged complete. This is the form in which requirements pass to the development team, and the form in which they are later tested.

RFQ 7 requires traceability matrices ensuring all requirements are addressed and validated throughout the project lifecycle. The matrix is established at MS1 and maintained as a live artefact through to handover. Each row traces a single requirement across its full path:

RFQ clause or stakeholder source → requirement identifier → design element → use case → build item → test case → acceptance evidence → deployment release.

The matrix is reviewed at each milestone. A requirement with no test case, or a test case with no acceptance evidence, is visible as a gap rather than discovered at UAT. The matrix is delivered to OIA as part of the handover pack and is the instrument by which OIA can verify that everything specified has in fact been built and validated.

RFQ 7 requires the Business Analyst to outline a risk assessment plan identifying and mitigating potential risks associated with the implementation. A risk register is established in week 2 and reviewed weekly. Each risk

carries an owner, a probability and impact assessment, a mitigation, a trigger that would escalate it, and a residual position. Risks span delivery, integration, data, security, adoption and compliance. The register distinguishes risks owned by us from those owned by OIA, and the latter are reflected in the dependency register within the project plan.

RFQ 7.5 requires the Business Analyst to ensure compliance with data governance, data security and privacy regulations and to apply data masking where required. The Business Analyst maintains a data inventory covering every category of personal data the solution handles, its lawful basis, its retention period, who may access it, and where masking applies. Non-production environments operate exclusively on masked data. The inventory is reviewed with OIA's compliance function before each release and forms part of the records required under the MTCIT AI policy.

RFQ 7 requires the Business Analyst to participate in testing activities including the creation of test cases and test plans. The Business Analyst authors the acceptance test cases directly from the requirements baseline, so that what is tested is what was specified. The Business Analyst facilitates UAT with OIA users, manages defect triage against the requirements, and signs the traceability matrix as complete before each release is proposed for sign-off.

RFQ 7.7 requires the Business Analyst to identify potential use cases beyond the initial implementation and propose scalable solutions for future enhancements. Because the solution is built on a shared business model rather than five separate applications, further use cases are additions to an existing foundation rather than new builds. During the engagement the Business Analyst maintains a forward register of candidate use cases observed during discovery but outside current scope — each recorded with its trigger, the outcome it would produce, the data it would require, and whether the current model already supports it. The register is delivered to OIA at handover as an input to future planning.

- **Stakeholder engagement plan** — Week 1
- **Current-state process model and system landscape** — Week 3
- **Gap analysis report** — Week 3
- **Risk register (established, then maintained weekly)** — Week 2
- **Requirements specification — functional, non-functional, user stories, use cases** — Week 4, signed off at MS1
- **Requirements traceability matrix (established, then maintained)** — Week 4
- **Data inventory and masking specification** — Week 4
- **Acceptance test cases and test plans** — Per release wave
- **Forward use-case register** — At handover

Anonymised historical recruitment data is used for calibration and fairness testing. The specification below sets the requirement unambiguously so that the delivery date can be placed on the project schedule.

Matching benchmark — Historical applications with the job description, the candidate profile as it stood, the actual shortlist or rejection outcome, and the stated reason — At least 300 applications spanning at least three distinct job families — Structured export; personal identifiers removed with record structure and relationships preserved — W4

Arabic tuning and validation — Gulf-dialect interview-style speech with transcripts, or Arabic and bilingual CVs — At least 10 hours of speech with transcripts, or at least 500 CVs — Audio in a standard lossless format with time-aligned transcripts; CVs in original file format — W4, second tranche by W8

Fairness testing — Historical decisions carrying gender and nationality labels — The same 300-application set, with labels supplied separately — Labels are used for testing only and never as model input — W4

Quality requirement. Records must be complete enough that the outcome is attributable; an application with no recorded decision or no stated reason cannot serve as a benchmark. A data quality assessment is performed within one week of receipt and coverage is reported against the requirement above.

Where data is incomplete or delayed. Each case has a defined fallback so that the dependency does not sit unmanaged on the critical path.

- **Matching benchmark unavailable or below the stated volume** — The benchmark set is constructed from synthetically generated symmetric cases reviewed and approved by OIA recruiters. The recall criterion is then stated against that agreed set rather than against historical outcomes
- **Arabic material unavailable** — Public Gulf-dialect corpora are used for the initial benchmark, with the accuracy figure stated against that corpus and revalidation against OIA data scheduled during the support period
- **Fairness labels unavailable** — Symmetric-pair testing proceeds on constructed pairs, which is the more rigorous method; historical labels serve as a secondary check
- **Delivery later than the dates above** — The affected activity moves by the period of the delay under the dependency register at §7.3, and the definition freeze for the benchmark set moves with it

In each case the commitment structure is preserved: the criterion is agreed, the measurement method is agreed, and where the input is weaker the criterion is stated against what is available rather than withdrawn.

UIS Cloud will track the following inputs in the dependency register, with an owner, required date, impact, workaround and escalation route. Mock services and representative synthetic data will keep development moving when external access is pending.

- Jadarah API, authentication, data dictionary, sandbox owner and delivery date.
- Confirmation of the design points for annual CV volume, live interview concurrency, assessment concurrency and campaign processing windows.
- Avatar class and the OIA-provided or commissioned image/rig/voice assets and usage rights.
- Target Arabic coverage: MSA, Omani dialect, general Gulf and expected code-switch patterns.
- Permission, consent and ownership for an OIA-specific speech test set and native transcription.
- Voice persona, gender/accent preference, ownership, permitted derivatives and withdrawal terms.
- Confirmation of the distribution, discipline mix, difficulty profile and psychometric scope for the committed 950 source items / 1,900 bilingual language versions.
- Classification of assessment tiers and the applicable browser-control, managed-client or OIA kiosk endpoint policy for each tier.
- Retention/deletion requirements for media, identity documents, transcripts, CVs, scores and audit evidence.
- OIA standards for Kubernetes, Linux, database, object storage, SIEM, KMS, backup and GPU vendor/SKU.
- Availability, RPO/RTO, maintenance windows and active-session continuity objectives.
- Permitted/prohibited licence categories and the commercial-component approval route.
- UIS Cloud's Oman support model, escalation hours and named operational responsibilities.



7.10 Project Management and Implementation Methodology (5%)

This section responds to Evaluation Criterion 7 and to the Project Management and Implementation Methodology requirements at RFQ section 8.

Delivery runs to OIA's project management approach, processes and templates, including the Project Charter and the status reporting format. The bidder completes the Project Charter to OIA's template at inception. Governance operates at three levels: a weekly delivery meeting on progress, risks and dependencies; a milestone review at each release gate; and a steering checkpoint at MS1, at go-live readiness and at handover.

The method is iterative within a fixed structure. Requirements are baselined once and changed only through change control; delivery proceeds in three release waves; each wave is verified against acceptance criteria defined before build begins. What is deliberately not iterative is the definition of done — the acceptance criteria are agreed at MS1 and do not move.

The activities below sit with OIA and are on the critical path. They are tracked in a dependency register reviewed at every weekly meeting. Late delivery of an input extends the affected activity by the period of the delay.

Quality Assurance Controls:

- **Acceptance criteria defined before build** — At MS1, per use case, with the measurement method stated alongside each criterion
- **Requirements traceability** — Maintained continuously; a requirement without a test case is visible as a gap before UAT, not at it
- **Automated verification** — Acceptance criteria instrumented in the build pipeline where measurable automatically; regression suite run on every build
- **Fairness verification** — Symmetric-pair test harness run before each release and periodically thereafter
- **Security verification** — Security testing in the pipeline; independent penetration test before go-live; all OIA assessment findings in the delivered solution closed before release
- **Release gate** — No release proceeds to production without acceptance criteria verified, traceability complete, and security findings closed to schedule

Changes to the requirements baseline follow a defined route: raised in writing, assessed for impact on scope, schedule and acceptance criteria, and approved by both parties before work proceeds. Where OIA issues an instruction under Article 3 of the contract terms, the instruction is carried out and the associated variation detail submitted within the seven calendar days the contract specifies. The change log is a standing item at the weekly meeting and is delivered with the handover pack.

Staffing profile by phase.

- **Discovery and definition** — W1–4 — 5–6 — Project Manager, Business Analyst, Solution Architect, Integration, Infrastructure
- **Wave 1 build** — W5–12 — 15–16 — Adds AI and NLP, front-end, identity, speech
- **Wave 2 and Wave 3 build, overlapping** — W9–21 — 18–20 — Adds assessment and proctoring, avatar, behavioural analysis, quality assurance, security
- **UAT, training and handover** — W22–26 — 8–10 — Project Manager, Business Analyst, quality assurance, technical writer, infrastructure, discipline leads on call

On-site presence in Oman. Attendance is planned for kick-off and the requirements workshops (W1–4), integration testing, each production deployment, user acceptance testing (W22–24) and training delivery (W24–26). The Business Analyst is present at OIA as required throughout the engagement. Local first-line support capability is established before go-live and continues through the support period.

Continuity. Key personnel are named and committed for the duration. Any substitution is made with equivalent or better qualifications and is notified to OIA in advance. Each discipline has an identified deputy so that delivery does not depend on a single individual.

- **OIA** — Business ownership, Jadarah/IdP access, representative data and consent, policy/retention decisions, infrastructure standards, UAT and final acceptance
- **UIS Cloud** — End-to-end delivery accountability covering Oman-based coordination, solution architecture, application and AI engineering, component selection, Jadarah integration, infrastructure and network readiness, security and operations alignment, Arabic stakeholder coordination, performance and quality validation, L1-L3 support, knowledge transfer and as-built documentation
- **OIA-UIS Cloud governance** — Architecture decisions, scope/change control, acceptance thresholds, risk escalation, go-live decision and post-launch model/change gates

7.11 Training and Handover (3%)

This section responds to Evaluation Criterion 8 and to the Training and Handover requirements at RFQ section 8.

Training is role-based and operationally oriented, delivered at OIA's premises, and covers both IT and non-IT audiences. Content, activities, timelines and facility requirements are set out below and confirmed at requirements sign-off.

- **Recruiters and HR end users** — Candidate ranking and evidence review; shortlist and rejection with recorded reasons; assessment configuration and results; interview configuration, evaluation review and outcome recommendation; reporting and export — Two half-day sessions per cohort, hands-on in the training environment
- **Super administrators** — Scoring weights and skill taxonomy, model parameters and maximum scores, retention policy, role assignment, fairness validation reporting, audit reconstruction — One full day
- **IT — server and application administration** — Architecture walkthrough, deployment and release process, rollback, monitoring and logging, backup and restore, high-availability operation and failover verification, capacity management — Two full days
- **IT — first-level support** — Common incident patterns, triage and escalation, diagnostic tooling, ticket handling against the SLA, known-issue register — One full day
- **Compliance** — Audit trail and decision reconstruction, fairness validation reports, data inventory and retention operation, regulatory reporting — One half-day session

A train-the-trainer session is offered for OIA staff who will onboard colleagues after handover, so that training capability remains with OIA.

- **User guide** — Simple and user-friendly, built primarily on screenshots with brief instructions, one guide per role, in English and Arabic
- **Administration and maintenance guide** — Operational procedures, configuration reference, troubleshooting, recovery, in English and Arabic
- **Short videos** — Task-length recordings for end users covering the common paths, in English and Arabic
- **Training decks and exercises** — Delivered per audience, with exercise data in the training environment

All training material is delivered in editable soft-copy format in English and Arabic, so that OIA may amend it as the system evolves. Every document carries a clear indexing scheme supporting search and cross-reference.

- **Documentation set** — High-level and low-level design, integration, development and as-built documents; maintenance and user guides; delivered in PDF, Word, Excel, Visio and Microsoft Project as suits the content, with a PowerPoint summary of the set
- **Source code and configuration** — Per Annex B, in a planned handover activity conducted with OIA's technical team, including a code walkthrough of the delivered application layer and the build and deployment path
- **Business data model** — The entities, rules, scoring dimensions, weights and decision-gate definitions, with the configuration interfaces through which OIA maintains them
- **Question banks** — Full item set with categorisation and review records, as OIA content
- **Test assets** — Test plans, test cases, the acceptance harness and the fairness test harness, so OIA can re-verify independently
- **Operational assets** — Runbooks, monitoring configuration, backup and restore procedures, known-issue register, dependency and change logs
- **Document register** — Index of every deliverable with version, date and owner

This annex defines the deliverable precisely, in response to General Requirement 4 and Article 10.3 of the contract terms. The solution is delivered in three layers, each with a stated ownership position. The dividing line is consistent throughout: everything carrying OIA's business meaning is delivered and owned; the general-purpose runtime beneath it is licensed.

- **Application logic** — All application-layer code for the five models
- **Business data model** — Entities, relationships, attributes, scoring dimensions, weights, thresholds and decision-gate definitions
- **Integration** — All Jadarah integration code — connectors, adapters, token validation, callback handlers and reconciliation logic
- **Interfaces** — Candidate, recruiter and administrative interfaces, including the accessibility implementation
- **Data structures** — Database schemas and migration scripts
- **Models** — Any model trained or fine-tuned on OIA data, with its training and evaluation scripts
- **Content** — Question-bank content authored for OIA
- **Deployment** — Deployment scripts, infrastructure-as-code and CI/CD pipeline configuration for the OIA environment
- **Test assets** — The acceptance criteria harness, the fairness test harness, and all test cases and test plans
- **Documentation** — High-level and low-level design, integration, development and as-built documents; maintenance and user guides; training materials
- **Operational data** — All data generated in operation — candidate profiles, session recordings, transcripts, evaluations and audit records — is OIA's throughout

These are handed over in a planned handover activity conducted with OIA's technical team, including a walkthrough of the delivered code and of the build and deployment path, so that OIA's team can build, deploy and modify the solution independently.

- **Reasoning engines** — The reusable computation engines on which use cases are composed
- **Agent orchestration framework** — The runtime that sequences and coordinates execution
- **Gate execution engine** — The mechanism that evaluates gates — OIA's gate definitions themselves are delivered under B.1
- **Data model runtime** — The mechanism that parses, validates and merges the business data model — OIA's own model content is delivered under B.1
- **Platform frameworks** — Observability, logging and access frameworks at platform level
- **Connector SDK** — The general connector development framework — the Jadarah implementation itself is delivered under B.1

These components predate this engagement, are developed independently of it, contain no OIA-specific content, and are deployed across multiple clients. They are licensed to OIA perpetually and irrevocably for on-premises operation, with no recurring licence fee and no dependency on an external activation service.

Source escrow. Source for these components is deposited with an independent escrow agent and released to OIA on any of the following: the supplier's insolvency or cessation of business; failure to meet support obligations beyond an agreed cure period; or discontinuation of the product line. The escrow agreement is executed at contract signature and the deposit verified annually.

- **Open-source components — speech recognition, speech synthesis, language models, supporting libraries** — Licensed under permissive terms. OIA may use, modify and redistribute them under those terms, with no dependency on the supplier. A complete bill of materials with licence identifiers is delivered with the handover pack
- **Commercial components, where used** — The bidder secures OIA's right to operate the component on its own infrastructure for the life of the solution, deployed offline with no online licence validation. Product edition and licence terms are itemised in the commercial submission

General Requirement 4 asks for the original source code to be handed over "if applicable". That qualifier distinguishes work created for OIA from a licensed product operated by OIA, and this annex applies it accordingly. Article 10.3 of the contract terms is consistent: the Contractor retains the copyrights it holds, while granting OIA an irrevocable, unconditional, unlimited, non-exclusive and royalty-free licence to use. Nothing in this annex reduces what OIA receives: the solution may be operated indefinitely without further payment, every element carrying business behaviour may be modified by OIA without reference to the supplier, and continuity is protected by escrow.

7.12 Sizing Information (5%)

This section responds to Evaluation Criterion 9 and to the Sizing Information at RFQ section 8. The reference hardware configuration is at Annex C.

- **Interviews** — 200 per month, stated in the RFQ — 10 concurrent sessions at peak
- **Assessments** — Not stated; OIA asked the bidder to propose an average — 800 per month average, rising to 2,500 during graduate-intake campaigns; 50 concurrent sessions at peak
- **Super administrators** — 5–10, stated in the RFQ; OIA only, confirmed in clarification — 10
- **Recruiter and hiring manager accounts** — Not stated — 50 named accounts, extensible without licence impact
- **Candidate accounts** — Managed in Jadarah — No account limit within the platform

- **Storage** — Session recordings dominate — Sized for 200 interviews per month at full video and audio retention, with the retention policy determining steady-state volume

The assessment volumes above are the bidder's stated assumption, recorded so that capacity and the design point are unambiguous. Sustained volumes materially above these figures are handled as a variation.

The workload has two distinct profiles and the environment is designed for both. CV processing is high-volume batch work that tolerates queuing and is scheduled away from interactive peaks. Interviews are low-volume but strictly real-time — a session cannot queue — and hold GPU capacity for their duration. Assessments sit between the two: concurrent but not latency-critical in the same way.

Capacity is therefore allocated so that real-time interview sessions have reserved headroom and never contend with batch parsing, while batch throughput uses the remaining capacity opportunistically.

The infrastructure is designed to accommodate increases in the number of interfaces, data volumes and additional use cases over time, and supports horizontal scaling to handle increasing load while maintaining performance. Application and integration tiers scale by adding nodes behind the load balancer. Inference capacity scales by adding GPU nodes to the pool; concurrency scales approximately linearly with GPU capacity for interview sessions and with worker count for batch processing. Storage scales independently of compute. No component requires redesign to reach several times the stated volumes.

The acceptance rate will be set from OIA's confirmed campaign window. As a sizing example, processing 10,000 three-page CVs in eight hours requires 62.5 pages/minute. Applying a two-times peak and failure allowance gives a planning target of approximately 125 pages/minute.

The physical design separates latency-sensitive interviews from batch document and assessment workloads so that campaign ingestion cannot exhaust live-service capacity. The roles and capacity assumptions below form the initial infrastructure design. Weeks 1-2 measurements will confirm exact SKUs.

The 2 x 141 GB H200 profiles will be evaluated separately. Per-device memory, total memory, bandwidth, interconnect, tensor-parallel behaviour, power and support lifecycle will be compared on the same frozen workload. Sizing will follow this test sequence:

- Freeze ASR, LLM, TTS, avatar, OCR and embedding versions, precision and serving settings.
- Measure single-session GPU memory, time to first token/audio, real-time factor and avatar frame rate.
- Run ten concurrent interviews while batch workloads remain isolated and retain 25-30% headroom.
- Fail one complete GPU node and repeat the contracted peak test.
- Run the full CV pipeline for 60 minutes at increasing worker counts.
- Record power, temperature, memory fragmentation, queue depth, errors and recovery.
- Convert measured demand into exact GPU/CPU/memory/storage/network SKU and lifecycle requirements.

N+1 means that one surviving GPU node can serve the full contracted real-time peak. It does not mean that two nodes together can just meet the peak. Storage is derived from media quality, annual volume, retention, legal hold, replicas, backup and growth rather than a standalone terabyte estimate.

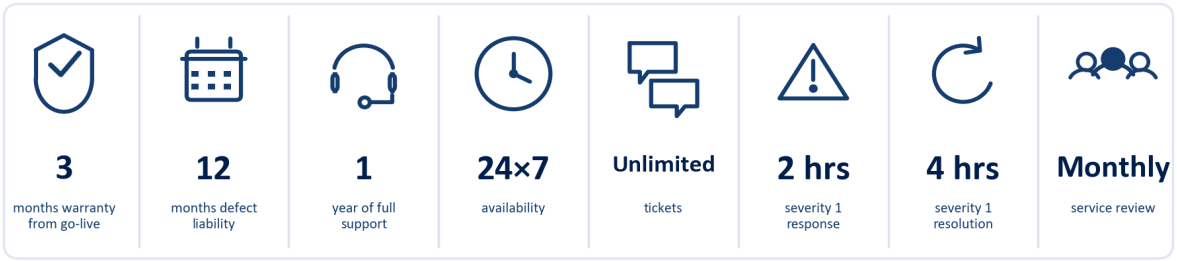
This annex specifies the infrastructure required to host and operate the solution, in response to System Requirement 10. All performance figures stated in this proposal are measured against this reference configuration.

- **Real-time AI** — Two independent GPU inference nodes in an N+1 configuration with pre-warmed serving; reference profiles include 4 x 48 GB or 2 x 80 GB GPUs per node — Each surviving node must

carry the contracted live-session peak; measured ASR, LLM, TTS, behavioural-analysis and avatar memory/latency confirm the exact SKU

- **Batch AI and document processing** — Isolated CPU/GPU worker pool for OCR, extraction, embeddings and offline scoring — Sustained complete-chain pages/minute, campaign window, queue behaviour and node-failure evidence
- **Application and control plane** — HA Kubernetes control plane with redundant application, workflow and integration workers — OIA platform standard, API load, maintenance and failure tests
- **Media edge** — Redundant OIA-hosted STUN/TURN and media gateway services — Concurrent interviews, weak-network tests, relay ratio and network security requirements
- **Data and evidence** — HA relational/vector data services, encrypted object storage, cache/queue, append-only audit and backup services — Retention, replication, RPO/RTO, media quality, legal hold, audit volume and growth
- **Network and facility** — Redundant switching; 25 GbE planning basis for GPU/data paths, subject to OIA standards — Exact GPU topology, storage throughput, rack power, cooling, support lifecycle and spare strategy
- **Interview sessions** — 200 per month; 10 concurrent at peak
- **CV processing** — Approximately 5,000 per year, confirmed in clarification — ≈420 per month sustained; campaign peaks to three times that rate absorbed through batch scheduling
- **Assessment sessions** — 800 per month average, 2,500 in campaign months; 50 concurrent at peak

The detailed infrastructure design is presented in the "AI Infrastructure Technical Proposal" document. It describes the proposed hosting model, system components, integration architecture, security controls, scalability, high availability, and d



eployment approach.

7.13 Maintenance and Support (5%)

This section responds to Evaluation Criterion 10 and to the Maintenance and Support requirements at RFQ section 8.

- **Warranty period** — Three months from go-live with full user support, as required by the Maintenance and Support scope
- **Defect liability** — Twelve months per Article 8.1.2 of the contract terms, accepted. Defects arising in that period are rectified at the bidder's cost. This is an obligation to rectify and is distinct from the warranty service period above
- **Support** — Comprehensive support for all delivered components for one year from the official launch date
- **Ticket volume** — Unlimited, as required
- **Availability** — 24×7
- **Account management** — A dedicated Account Manager handles day-to-day queries, support and complaints, and holds a monthly service review with OIA

- **Local capability** — First-line support operates in the Oman time zone, with escalation to specialist second and third line
- **Launch support** — An expert team is available on site as required during project launch, providing immediate response and rectification

The severity definitions and times below are as specified in the RFQ and are accepted in full.

- **1** — Complete system unavailability, system crash, or critical services unavailable — 2 hours — 4 hours
- **2** — Partial system unavailability; more than 50% of units unavailable — 2 hours — 24 hours
- **3** — Major functionality malfunction affecting less than 50% of units — 24 hours — 48 hours
- **4** — Minor functionality malfunction affecting one to two units — 24 hours — 7 days

A defect introduced into existing functionality by an enhancement is classified by its impact and treated on priority, in line with General Requirement 3. Severity is assigned on impact, not on origin.

- **Health check** — Monthly — Platform health, capacity trend, error and latency profile, backup verification, failover verification
- **Security patching** — On release, and immediately for critical vulnerabilities — Timely bug fixes and patching for security vulnerabilities across all delivered components
- **Functional updates** — Quarterly release train, or as agreed — Regular updates improving accuracy and functionality; configuration changes; new features and enhancements as agreed
- **Service review** — Monthly — Ticket profile, SLA performance, open issues, change pipeline, capacity outlook

A single point of intake receives incidents and requests through the agreed channels, with automatic acknowledgement and severity classification confirmed with the reporter. First line resolves known patterns; second line covers application and integration; third line covers AI components and platform. The Account Manager owns the relationship and escalation, and is accountable for SLA performance reported monthly. A known-issue register and knowledge base are maintained and handed to OIA so that first-line capability transfers over time.

7.14 Timeline

- **Weeks 1-2** configuration and verification sprint: During Weeks 1-2, UIS Cloud will confirm the project measurements and production configuration. This work is part of the six-month programme and does not change the scope or schedule. Outputs are:
    - Production component register and preliminary SBOM, Model BOM and asset registers.
    - Separate Arabic and English CV accuracy results and a sustained pages/minute test.
    - Native-streaming, multilingual and Arabic-specialised Gulf/Omani ASR comparison, including code-switching.
    - Arabic TTS/diacritisation/pronunciation-lexicon blind evaluation.
    - Single-session end-to-end trace, multimodal behavioural-analysis demonstration and avatar/fallback demonstration.
    - Initial GPU memory/latency profile and a test plan for ten-session/N+1 validation.
    - Signed register of measured results, acceptance targets, assumptions and dependencies.

    Measured results will state the dataset, software version, hardware profile and test method. Production-equivalent end-to-end and failure tests will provide the signed acceptance records for capacity, latency, active-session continuity and exact hardware configuration.

Period	Primary workstream	Exit evidence
Weeks 1-4	Weeks 1-2 configuration and verification sprint, platform foundation, canonical data model, Jadarah contract and DPIA	Production configuration, I1-I5 contracts, data dictionary, IdP, datasets/assets and signed test methods
Weeks 3-10	CV parsing plus matching/recommendation	Bilingual extraction results, source traceability, confirmation workflow and initial integration
Weeks 5-17	Assessment	Agreed bank scope, authoring, delivery, scoring, proctoring and psychometric review
Weeks 5-20	Speech, interview and avatar	Gulf/Omani ASR, TTS, dialogue, latency, sandbox, avatar and text fallback evidence
Weeks 11-22	Security, accessibility, fairness, HA/load and Jadarah E2E	Production-equivalent performance, failure, air-gap, security and WCAG reports
Weeks 23-26	UAT, remediation, cutover, training and handover	Go-live readiness, restore rehearsal, final BOM, as-built documents and operating handover

7.15 Conclusion

UIS Cloud will take end-to-end responsibility for an on-premise bilingual platform that covers the full recruitment workflow and keeps Jadarah as OIA's system of record. The design measures Arabic and Omani performance separately, preserves interviews through node failure, and keeps automated assessments reviewable by authorised OIA users.

The six-month programme includes configuration, integration, security review, performance and resilience testing, UAT and handover. The architecture, technology choices, capacity method and verification gates in this annex provide a clear basis for technical evaluation and contract acceptance.